



Graphing with ggplot2

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Math 241

Week 2 | Fall 2025

Announcements

- Office Hours Schedule is now posted on the course website.
 - We will try to stick to this schedule, but if one of us has to reschedule, we will send a Slack message and update the schedule.
- P-Set 0 due at 9am on **WEDNESDAY** (all other p-sets will be due on Thursdays).

Week 2 Goals

Mon Lecture

- Basics of `ggplot2`
- Explore several `geoms`.
- And a little data wrangling with `dplyr` as needed!

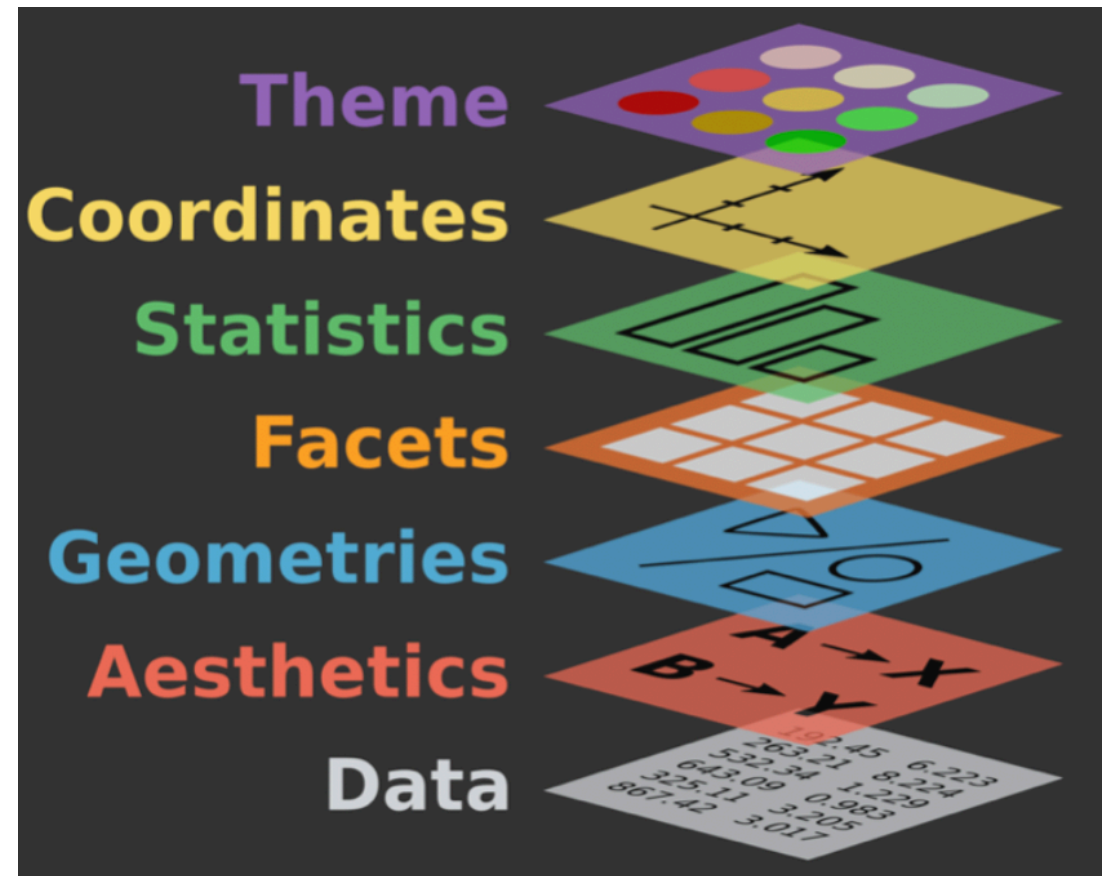
Wed Lecture

- GitHub workflow overview
- Graphing context!
 - Labels
 - Highlighting
 - Useful text
- Look at more `geoms`.
- Explore further customizations.
 - Color
 - Themes
- Learn how to ask coding questions well.



Recall: The Grammar of Graphics

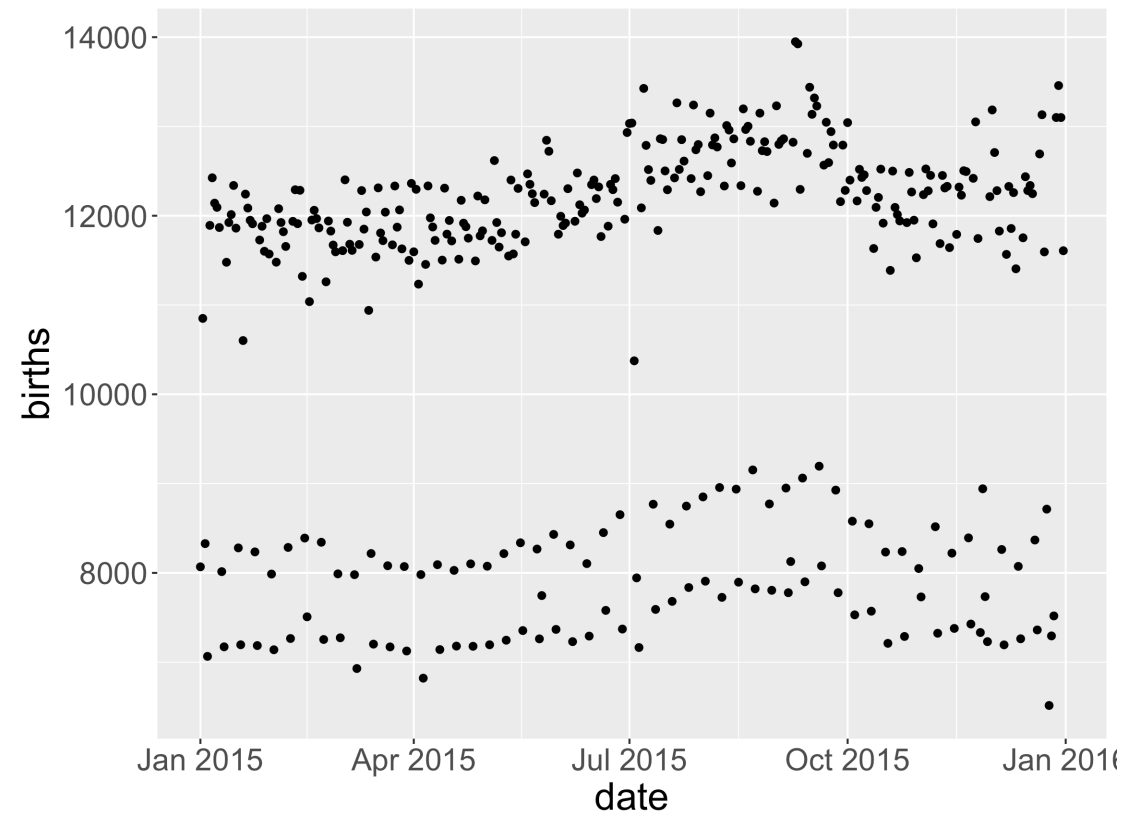
- **data**: dataset that contains the data
- **geom**: geometric shape that the data are mapped to
 - point, line, bar, text, ...
- **aesthetic**: visual properties of the **geom**
 - x position, y position, color, fill, shape
- **coord**: coordinate system
 - Cartesian, polar, geographic
- **scale**: controls how data are mapped to the visual values of the aesthetic
 - EX: particular colors, linear
- **guide**: legend to help user convert visual display back to the data



ggplot2 example code

```
1 ggplot(data = ---, mapping = aes(---)) +  
2   geom_---(---) +  
3   coord_---() +  
4   scale_---_---() +  
5   ---
```

Example: Over the course of a year, how does the daily number of births vary?



- What patterns do you see?



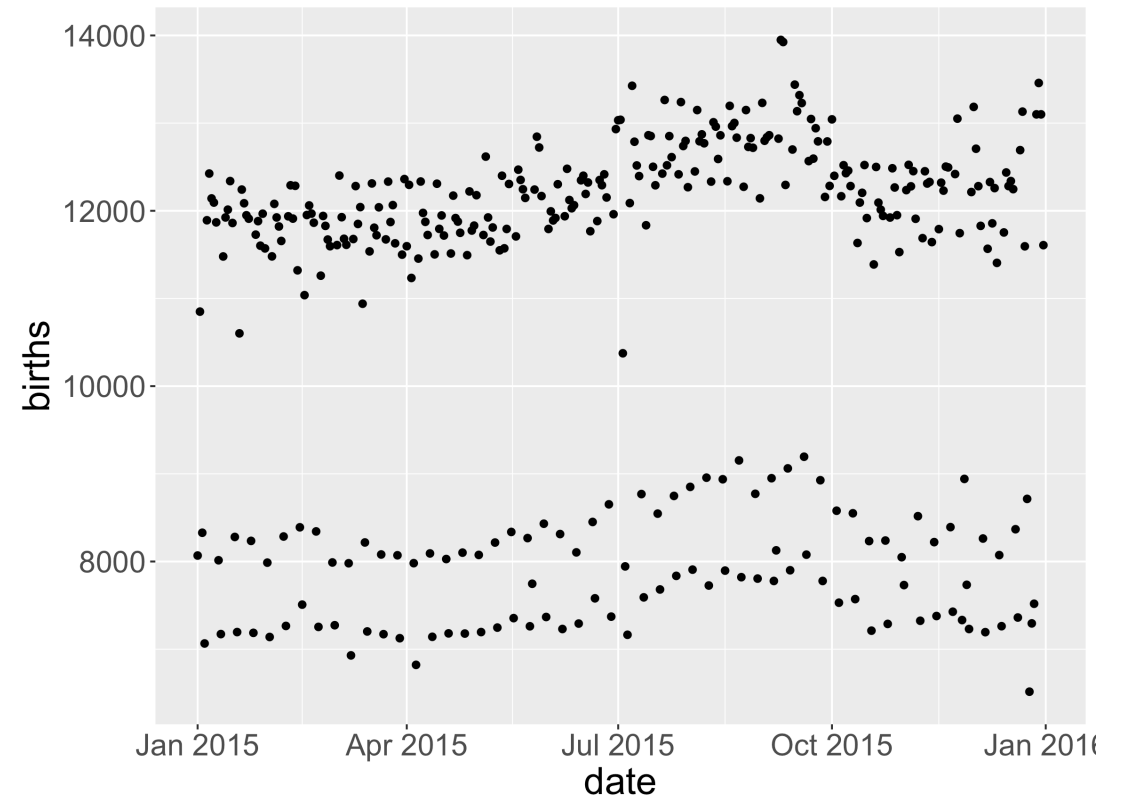
Example

```
1 # Load library that has dataset of interest
2 library(mosaicData)
3
4 # Grab data
5 data(Births2015)
6
7 # Load tidyverse (which contains ggplot2)
8 library(tidyverse)
```

Example

```
1 # Example code
2 ggplot(data = ---, mapping = aes(---)) +
3   geom_---(---) +
4   coord_---() +
5   scale_---_---() +
6   ---
```

```
1 # Create plot
2 ggplot(data = Births2015,
3       mapping = aes(x = date, y = births)) +
4   geom_point()
```



Example: Cycles related to day of the week?

```
1 # Look at structure of data with dplyr function
2 glimpse(Births2015)
```

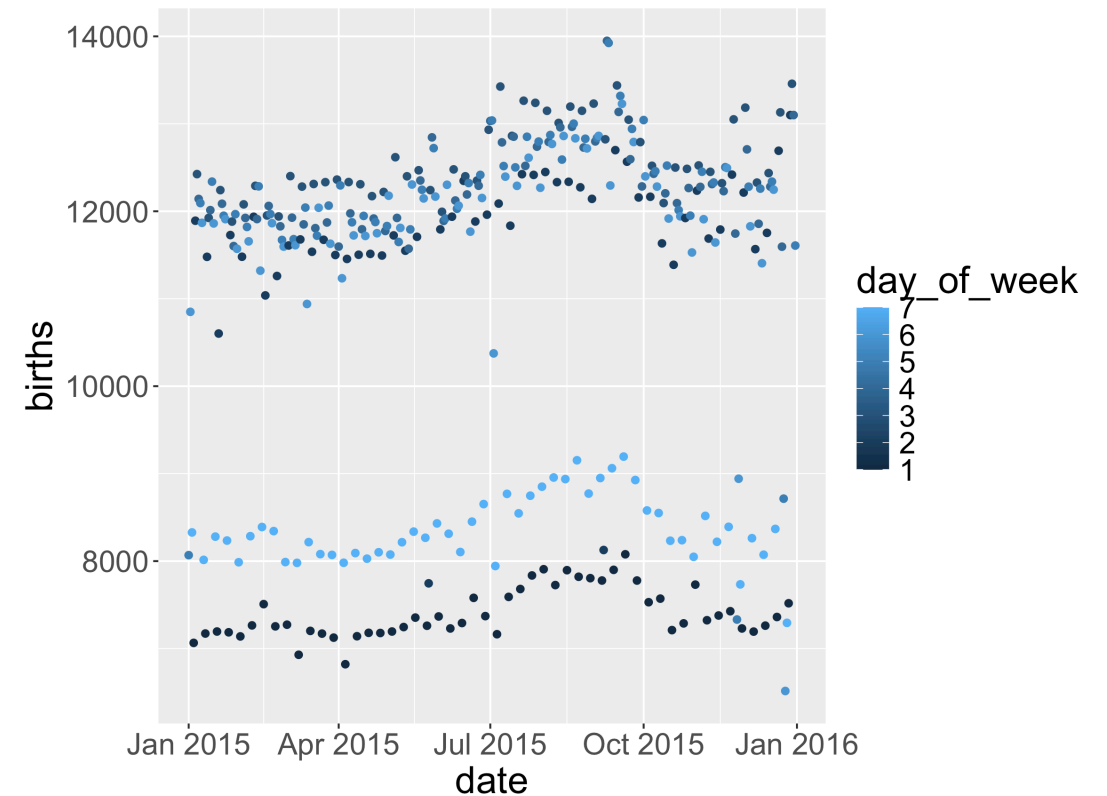
Rows: 365

Columns: 8

```
$ date      <date> 2015-01-01, 2015-01-02, 2015-01-03, 2015-01-04, 2015-01-...
$ births    <dbl> 8068, 10850, 8328, 7065, 11892, 12425, 12141, 12094, 1186...
$ wday      <ord> Thu, Fri, Sat, Sun, Mon, Tue, Wed, Thu, Fri, Sat, Sun, Mo...
$ year      <dbl> 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 2015, 201...
$ month     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ day_of_year <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17...
$ day_of_month <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17...
$ day_of_week <dbl> 5, 6, 7, 1, 2, 3, 4, 5, 6, 7, 1, 2, 3, 4, 5, 6, 7, 1, 2, ...
```

Example: Cycles related to day of the week?

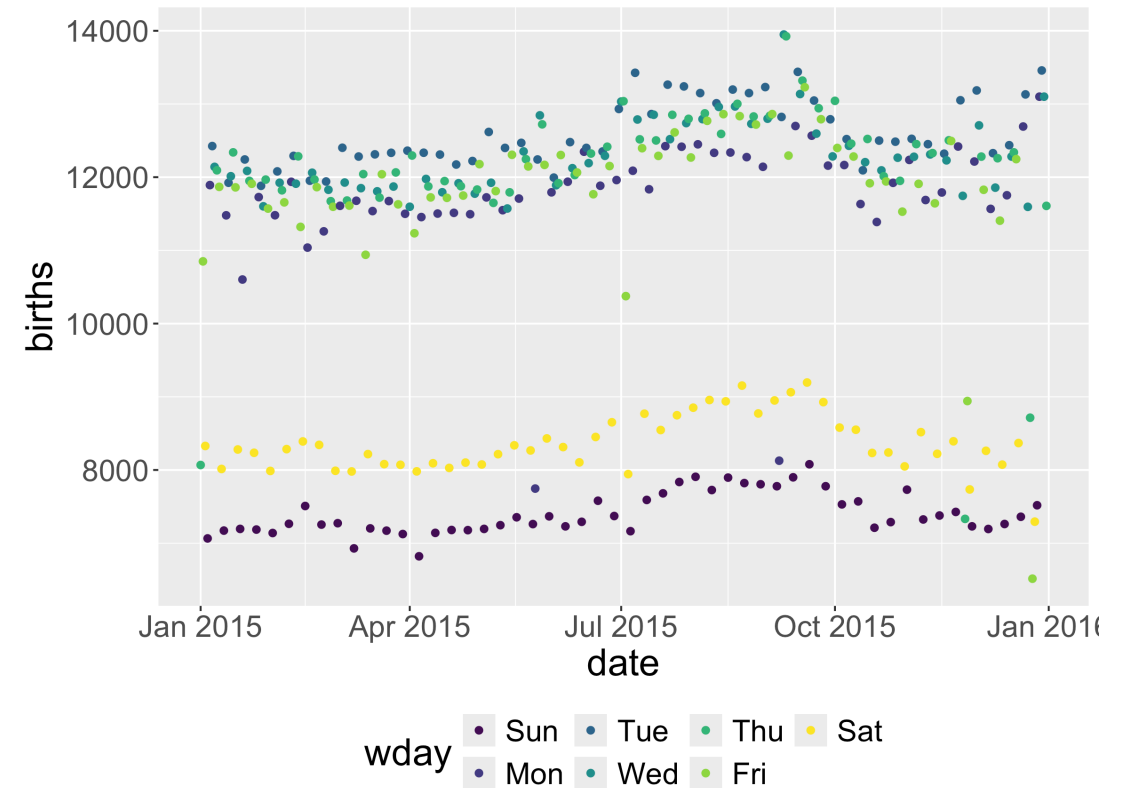
```
1 # Create plot
2 ggplot(data = Births2015,
3       mapping = aes(x = date, y = births,
4                     color = day_of_week)) +
5   geom_point()
```



- Additional aesthetic:
 - Why is this not quite what we want? What do we want?
- What happened to the aspect ratio when we added the legend?

Example: Cycles related to day of the week?

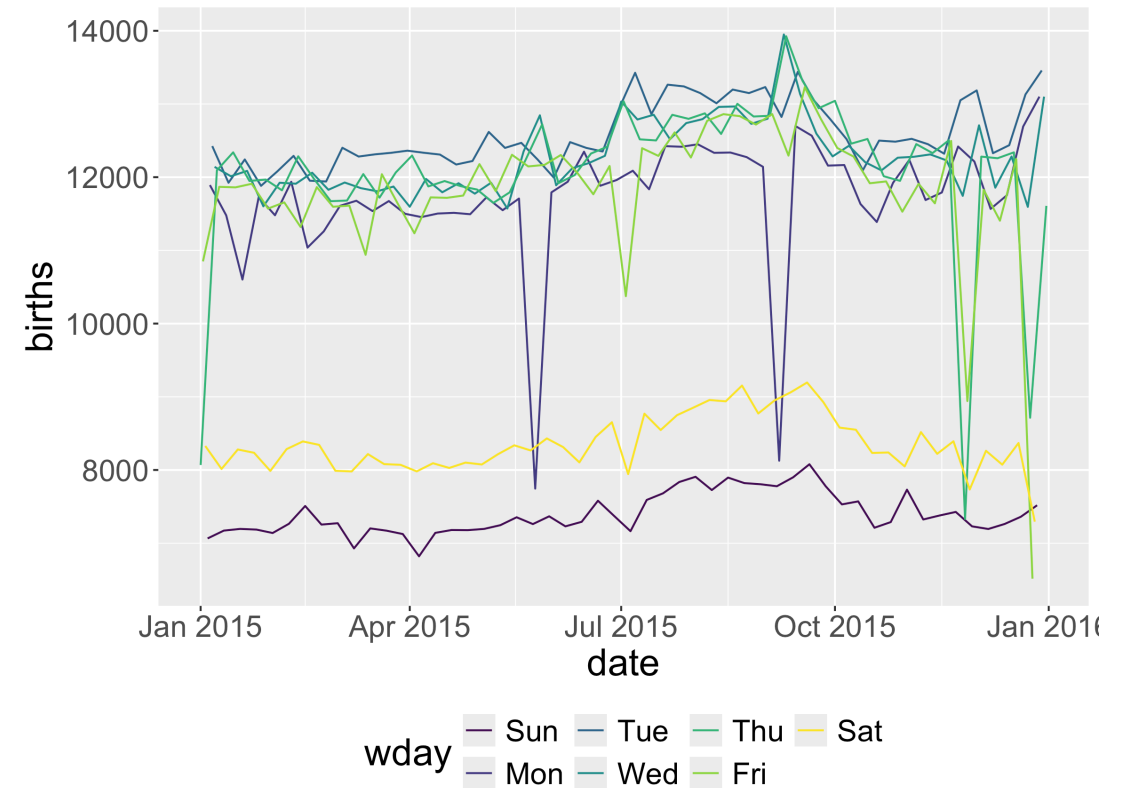
```
1 # Create plot
2 ggplot(data = Births2015,
3       mapping = aes(x = date, y = births,
4                     color = wday)) +
5   geom_point() +
6   theme(legend.position = "bottom")
```



- Positioning the legend on the bottom of the graph helped give us a nicer aspect ratio.
- What if we want to see the **direction** that the number of births take over time for each day of the week?
 - New visual cue/**geom**?

Example

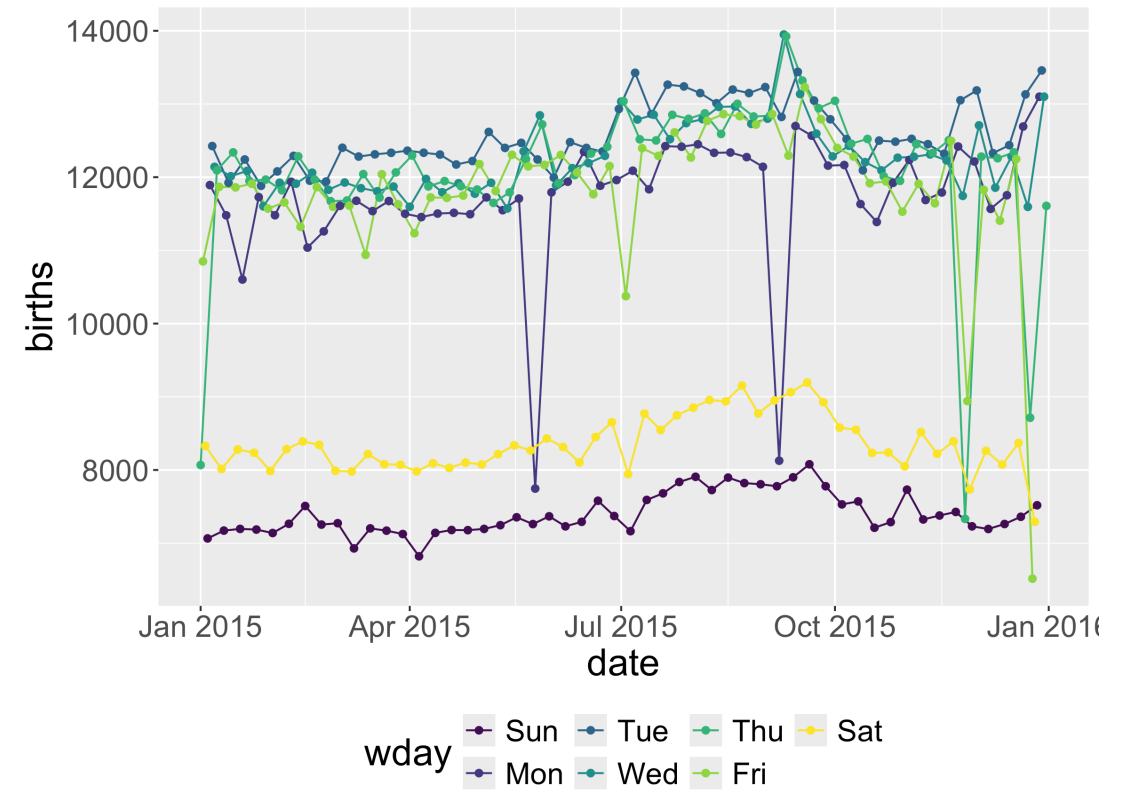
```
1 # Create plot
2 ggplot(data = Births2015,
3       mapping = aes(x = date, y = births,
4                     color = wday)) +
5   geom_line() +
6   theme(legend.position = "bottom")
```



- What if we want visual cues for both **position** and **direction**?

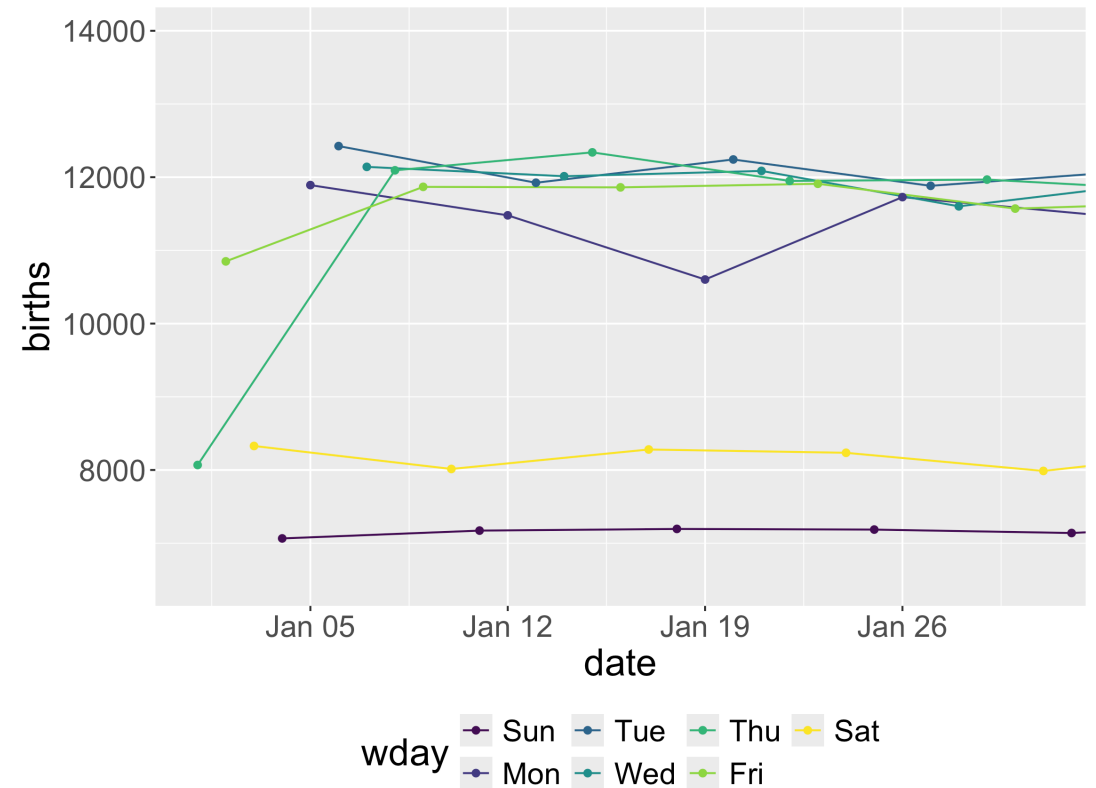
Example

```
1 # Create plot
2 ggplot(data = Births2015,
3       mapping = aes(x = date, y = births,
4                     color = wday)) +
5   geom_line() +
6   geom_point() +
7   theme(legend.position = "bottom")
```



Coordinate System Layer

```
1 library(lubridate)
2
3 ggplot(data = Births2015,
4       mapping = aes(x = date, y = births,
5                     color = wday)) +
6   geom_line() +
7   geom_point() +
8   theme(legend.position = "bottom") +
9   coord_cartesian(xlim =
10                  as_date(c("2015-01-01",
11                            "2015-01-31")))
```

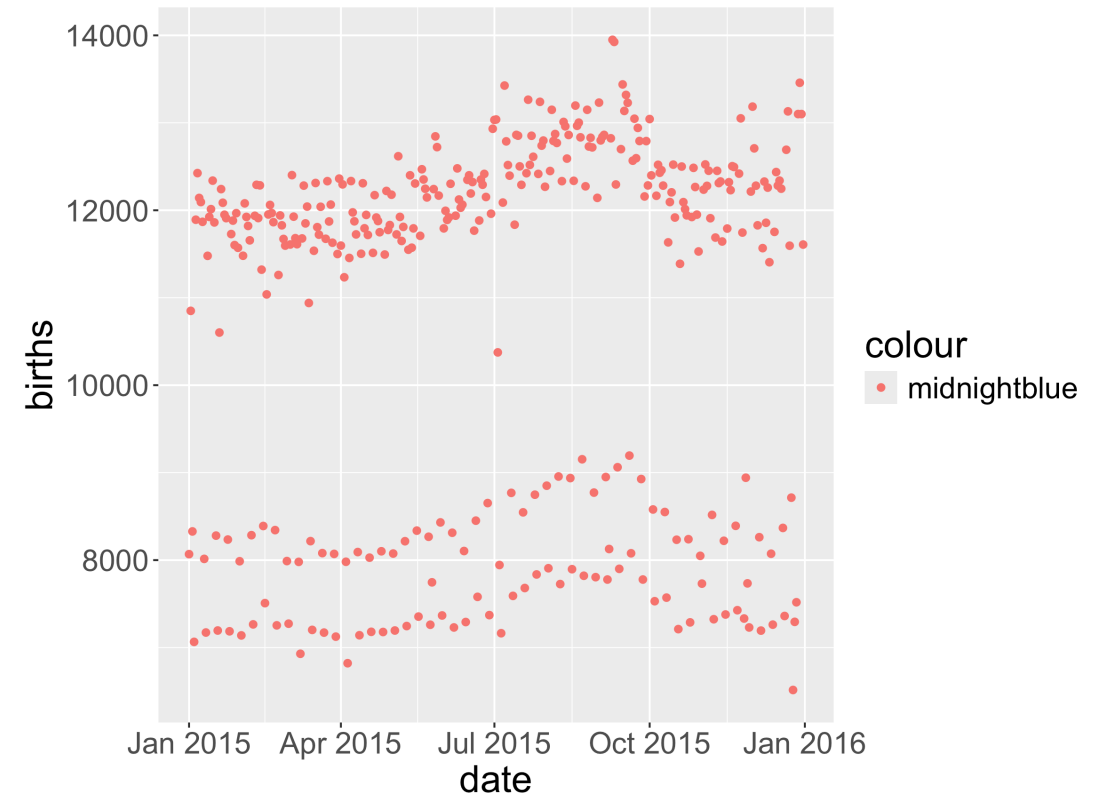


- How did this new layer change our plot?
- What if we want all the points to be colored “midnightblue”?



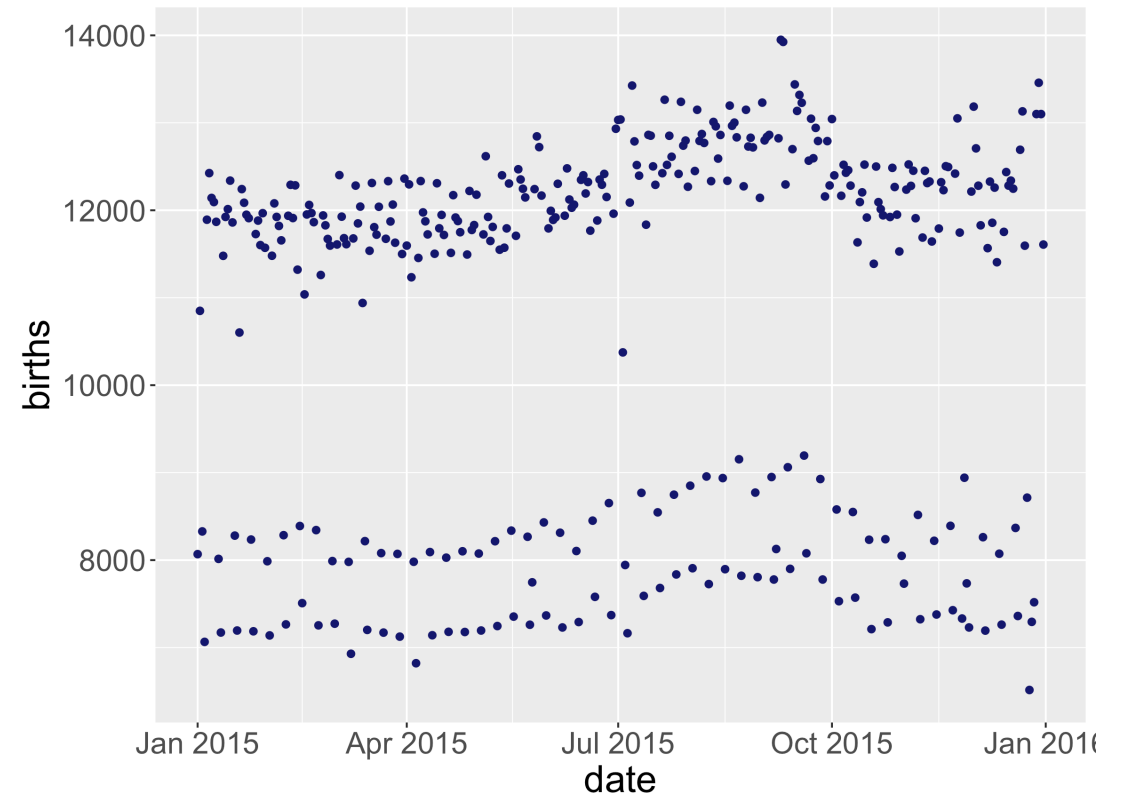
Setting instead of Mapping

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = "midnightblue")) +  
4   geom_point()
```



Setting instead of Mapping

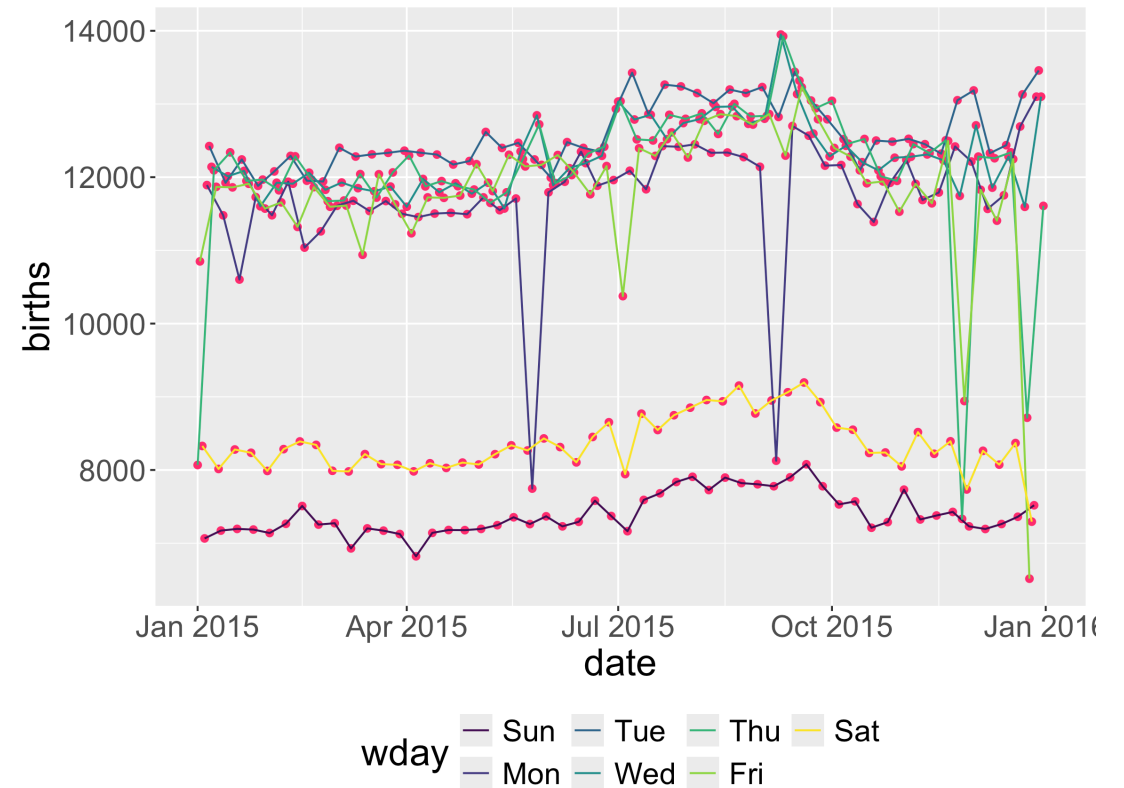
```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births)) +  
3   geom_point(color = "midnightblue")
```



- If you want to **set** an aesthetic to a specific value (instead of **mapping** the aesthetic to a variable), do so in the **geom_--()** function.

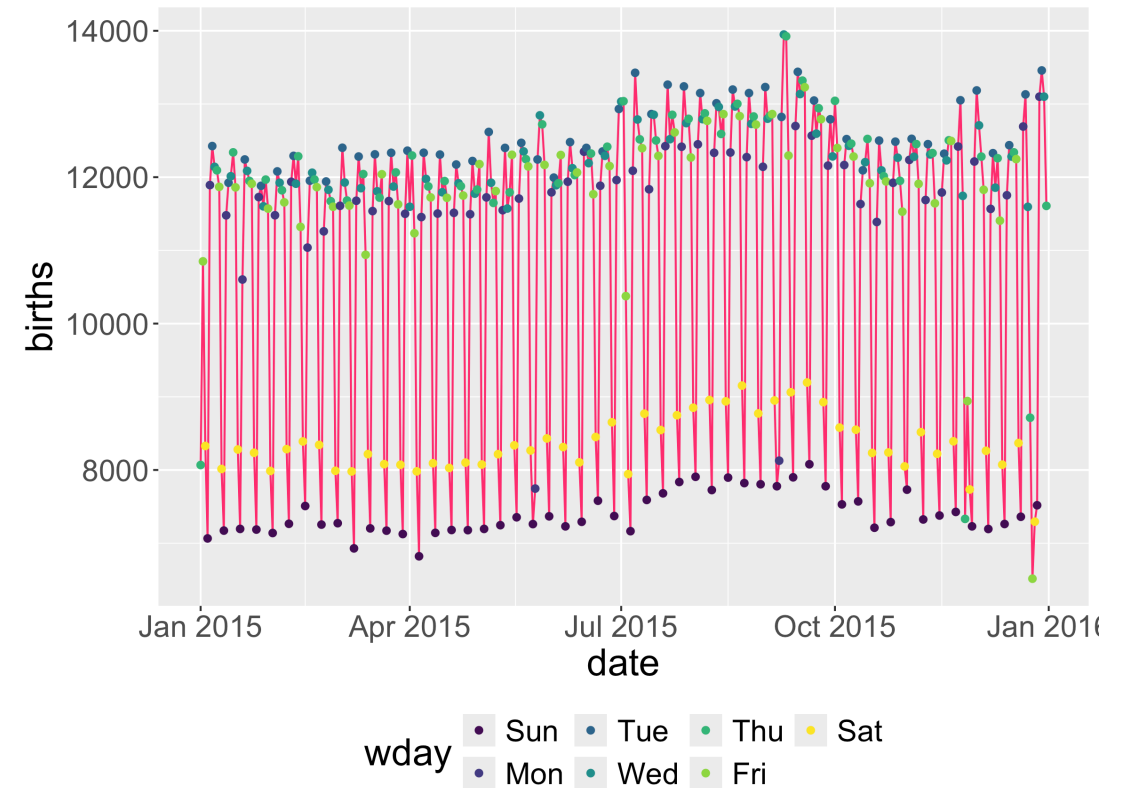
Layer order (sometimes) matters

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point(color = "#ff006e") +  
5   geom_line() +  
6   theme(legend.position = "bottom")
```



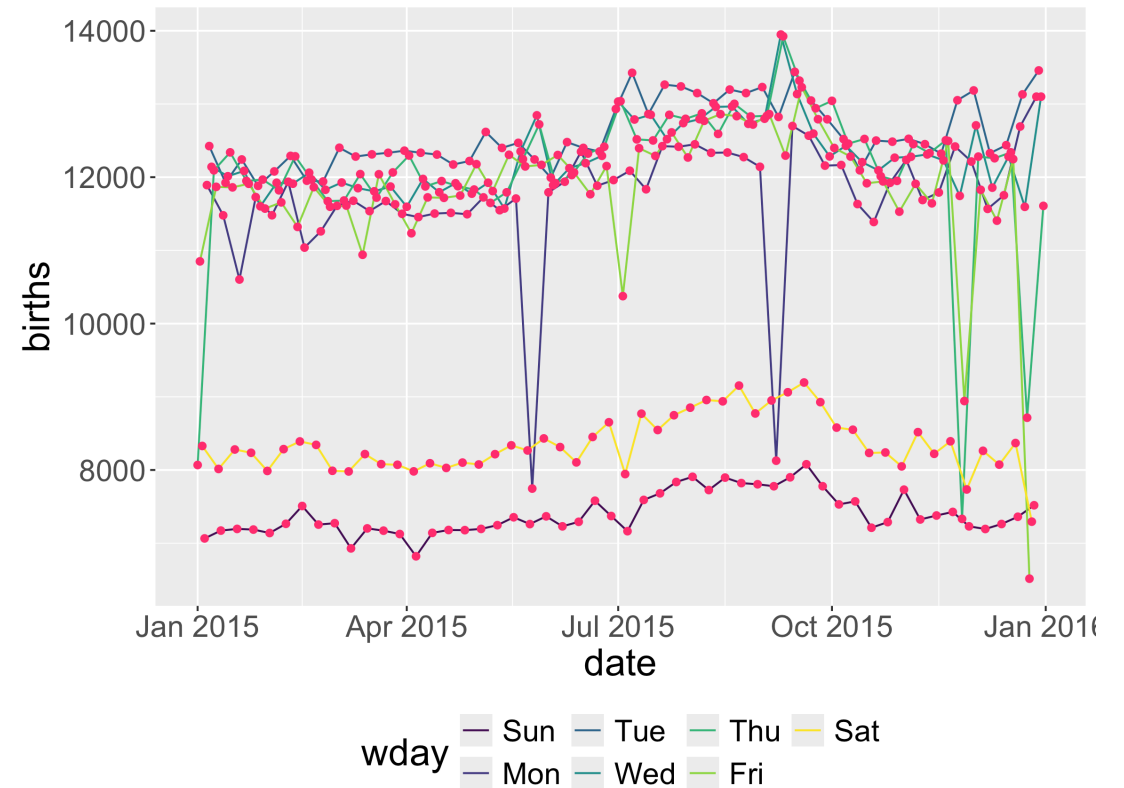
Layer order (sometimes) matters

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_line(color = "#ff006e") +  
5   geom_point() +  
6   theme(legend.position = "bottom")
```



Layer order (sometimes) matters

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_line() +  
5   geom_point(color = "#ff006e") +  
6   theme(legend.position = "bottom")
```



- Inheriting aesthetics discussion.

Let's explore other geoms

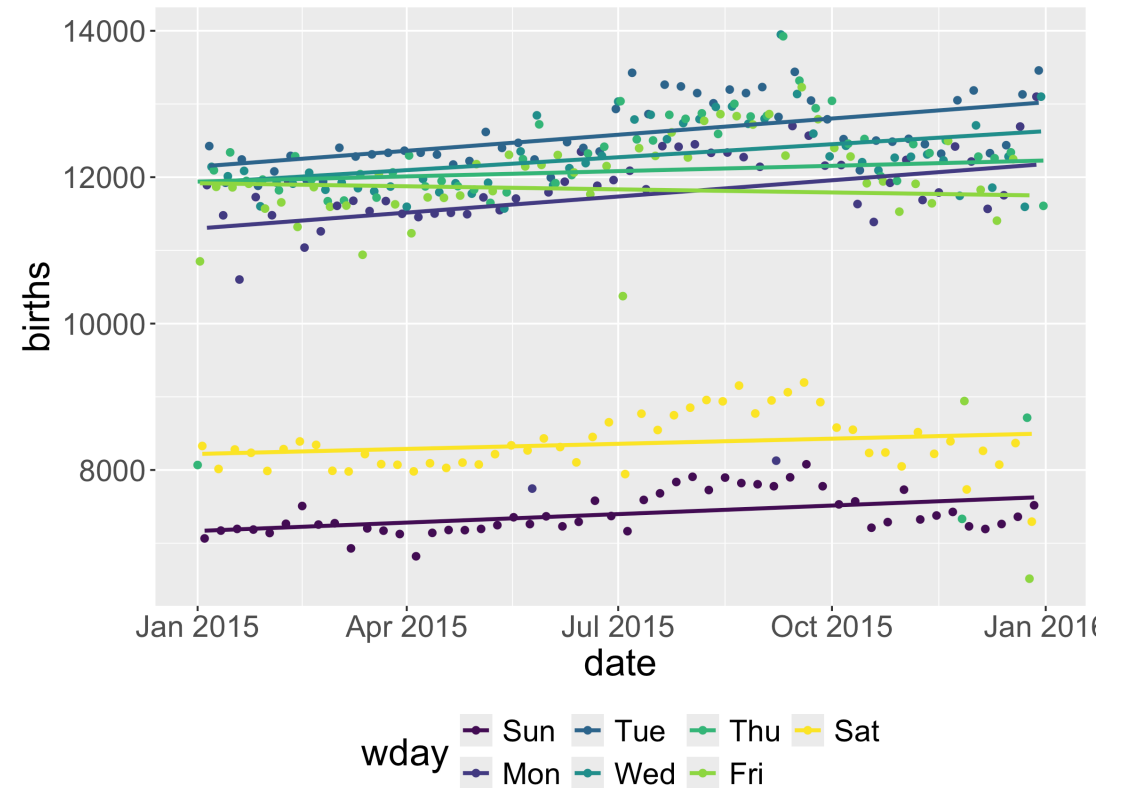
- Many are listed on the first page of the [ggplot2 cheatsheet](#).
- Can also ask R:

```
1 apropos("geom_")
```

```
[1] "geom_abline"      "geom_area"      "geom_bar"
[4] "geom_bin_2d"      "geom_bin2d"     "geom_blank"
[7] "geom_boxplot"     "geom_col"       "geom_contour"
[10] "geom_contour_filled" "geom_count"    "geom_crossbar"
[13] "geom_curve"       "geom_density"   "geom_density_2d"
[16] "geom_density_2d_filled" "geom_density2d" "geom_density2d_filled"
[19] "geom_dotplot"     "geom_errorbar"  "geom_errorbarh"
[22] "geom_freqpoly"    "geom_function"  "geom_hex"
[25] "geom_histogram"   "geom_hline"     "geom_jitter"
[28] "geom_label"       "geom_line"      "geom_linerange"
[31] "geom_map"         "geom_path"      "geom_point"
[34] "geom_pointrange"  "geom_polygon"   "geom_qq"
[37] "geom_qq_line"     "geom_quantile"  "geom_raster"
[40] "geom_rect"        "geom_ribbon"    "geom_rug"
```

Adding Curve(s)

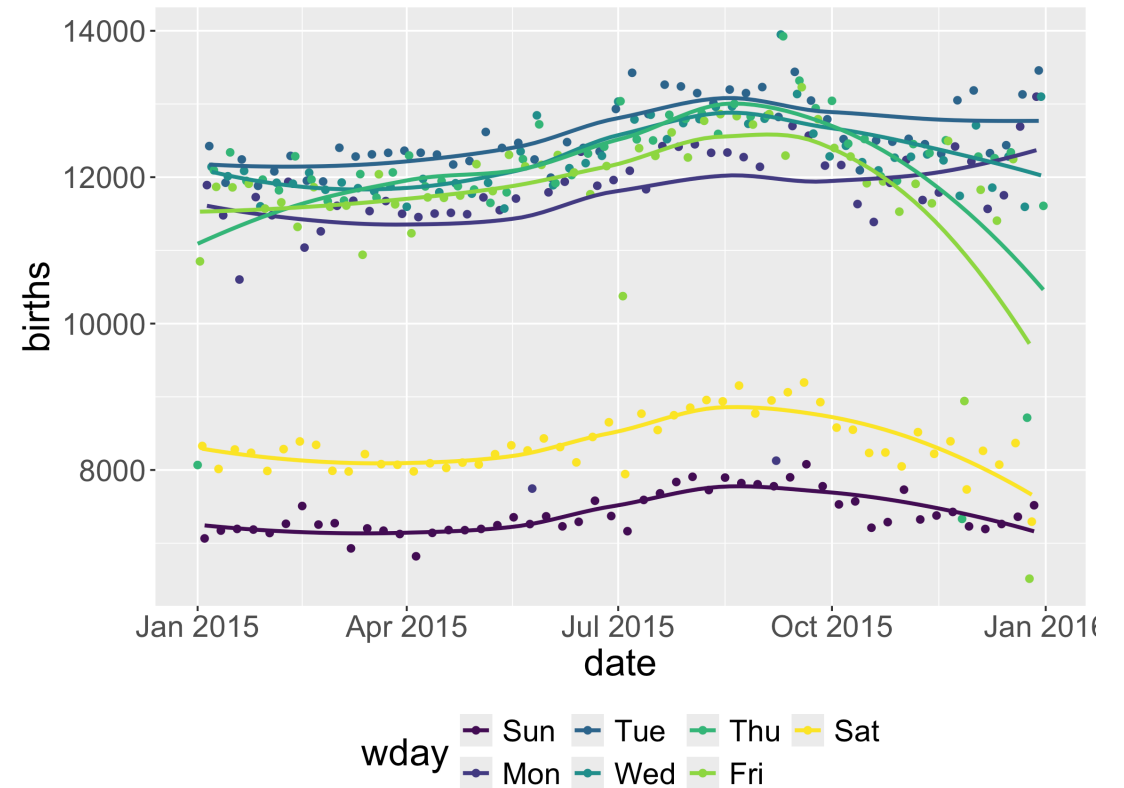
```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   geom_smooth(method = "lm", se = FALSE) +  
6   theme(legend.position = "bottom")
```



- Does a multiple linear regression line(s) capture the trend?

Adding Curve(s)

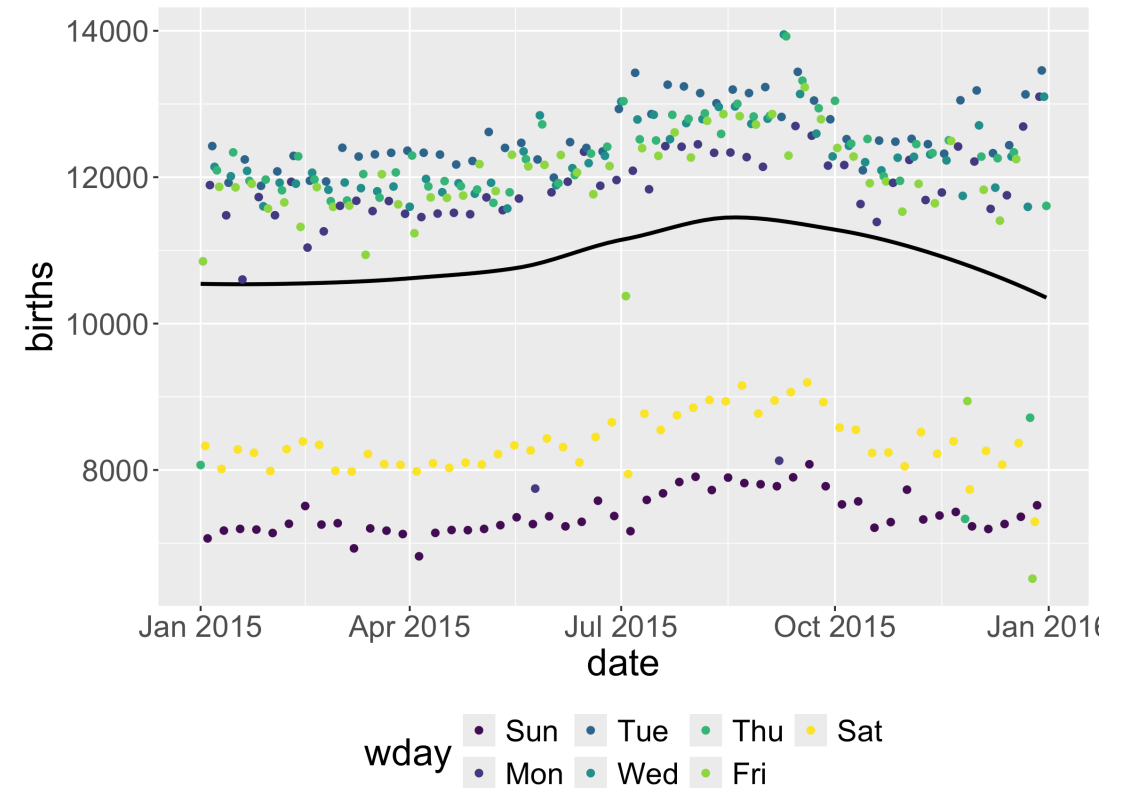
```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_point() +  
5   geom_smooth(se = FALSE) +  
6   theme(legend.position = "bottom")
```



- The default LOESS smoother usually does a reasonable job.

Adding Curve(s)

```
1 ggplot(data = Births2015,  
2       mapping = aes(x = date, y = births,  
3                     color = wday)) +  
4   geom_smooth(color = "black", se = FALSE) +  
5   geom_point() +  
6   theme(legend.position = "bottom")
```



- What happened?
- Inheriting aesthetics discussion.

New Example: Movies and the Bechdel Test

- Need a new dataset with more categorical variables
- **The Alison Bechdel Test:** A movie passes the test if:
 - There are at least two named women in the picture
 - They have a conversation with each other at some point
 - That conversation isn't about a male character
- Movies from 1970 - 2013

```
1 movies <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2021/2021-03-09,  
2   filter(rated %in% c("R", "PG-13", "PG", "G"))
```


New Example: Movies and the Bechdel Test

```
1 glimpse(movies)
```

Rows: 1,549

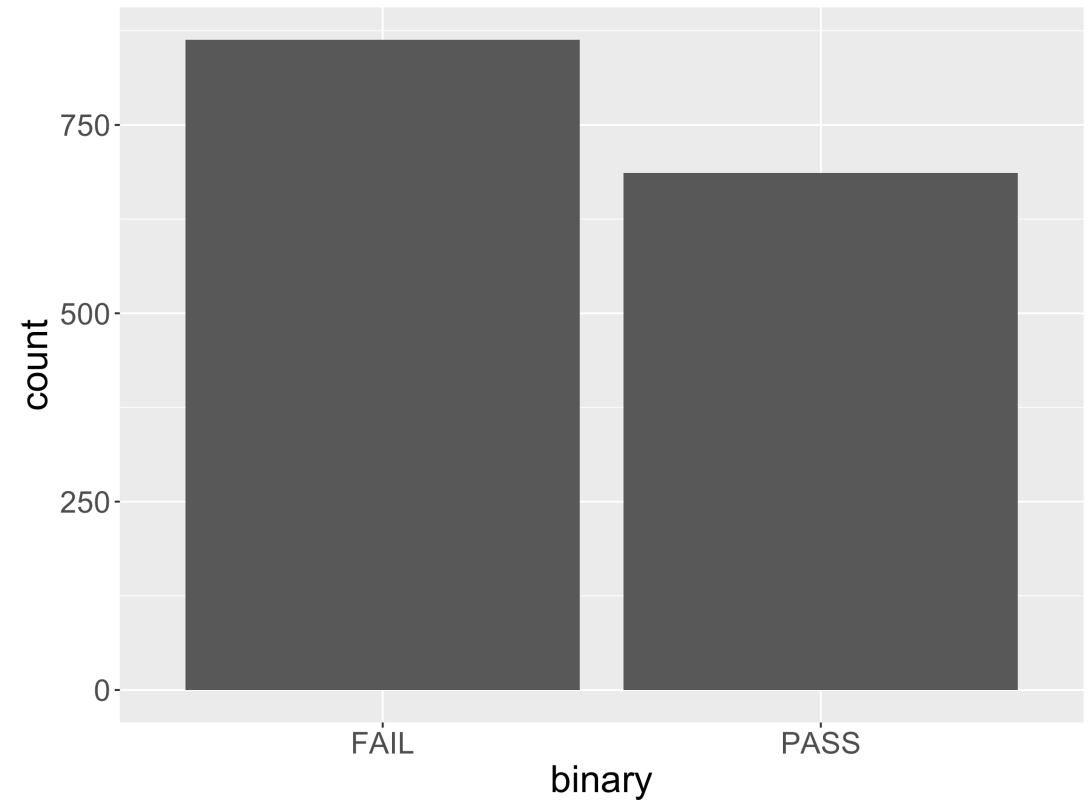
Columns: 34

```
$ year      <dbl> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 20...
$ imdb      <chr> "tt2024544", "tt1272878", "tt0453562", "tt1335975", "tt1...
$ title     <chr> "12 Years a Slave", "2 Guns", "42", "47 Ronin", "A Good ...
$ test      <chr> "notalk-disagree", "notalk", "men", "men", "notalk", "ok...
$ clean_test <chr> "notalk", "notalk", "men", "men", "notalk", "ok", "ok", ...
$ binary    <chr> "FAIL", "FAIL", "FAIL", "FAIL", "FAIL", "PASS", "PASS", ...
$ budget    <dbl> 2.00e+07, 6.10e+07, 4.00e+07, 2.25e+08, 9.20e+07, 1.20e+...
$ domgross  <chr> "53107035", "75612460", "95020213", "38362475", "6734919...
$ intgross  <chr> "158607035", "132493015", "95020213", "145803842", "3042...
$ code      <chr> "2013FAIL", "2013FAIL", "2013FAIL", "2013FAIL", "2013FAI...
$ budget_2013 <dbl> 2.00e+07, 6.10e+07, 4.00e+07, 2.25e+08, 9.20e+07, 1.20e+...
$ domgross_2013 <chr> "53107035", "75612460", "95020213", "38362475", "6734919...
```

What are useful **geoms** for describing
amounts/frequencies?

Amounts: `geom_bar`

```
1 ggplot(data = movies,  
2       mapping = aes(x = binary)) +  
3   geom_bar()
```



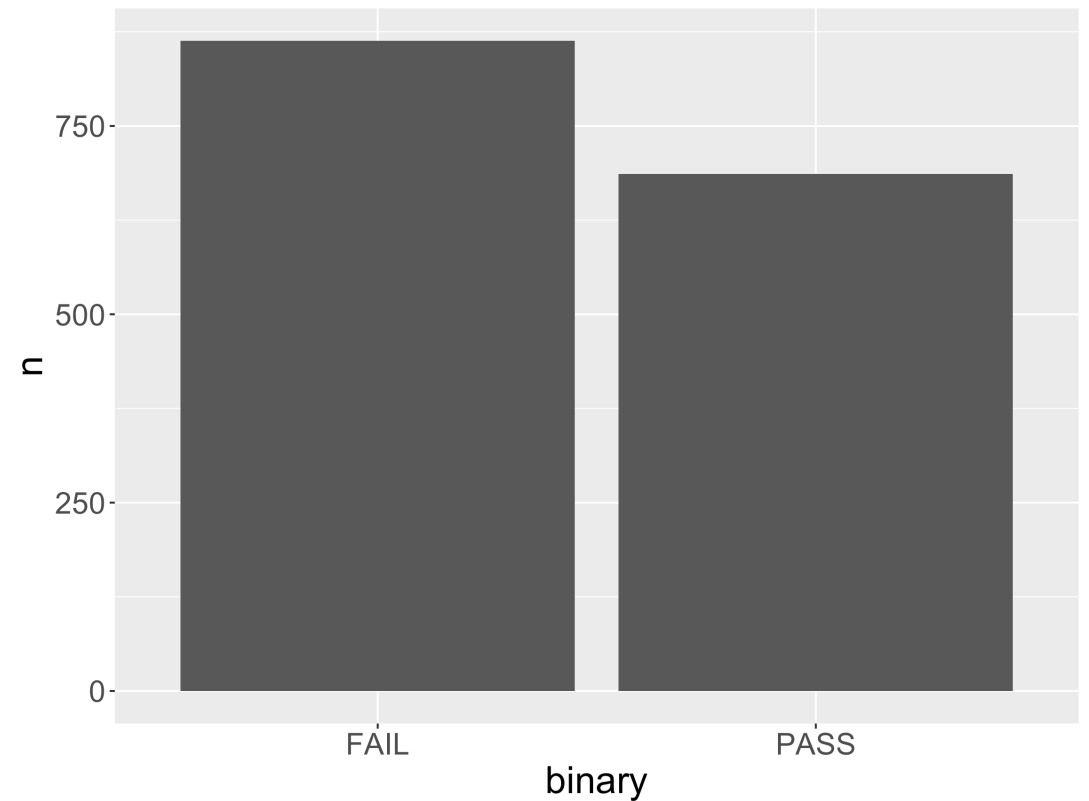
- Verbalize the mapping of `data` to `geom_bar()`.
 - How is this mapping different from `geom_point()`?

Another option: **geom_col**

```
1 # First wrangle with dplyr
2 movies_ag <- count(movies, binary)
3 movies_ag
```

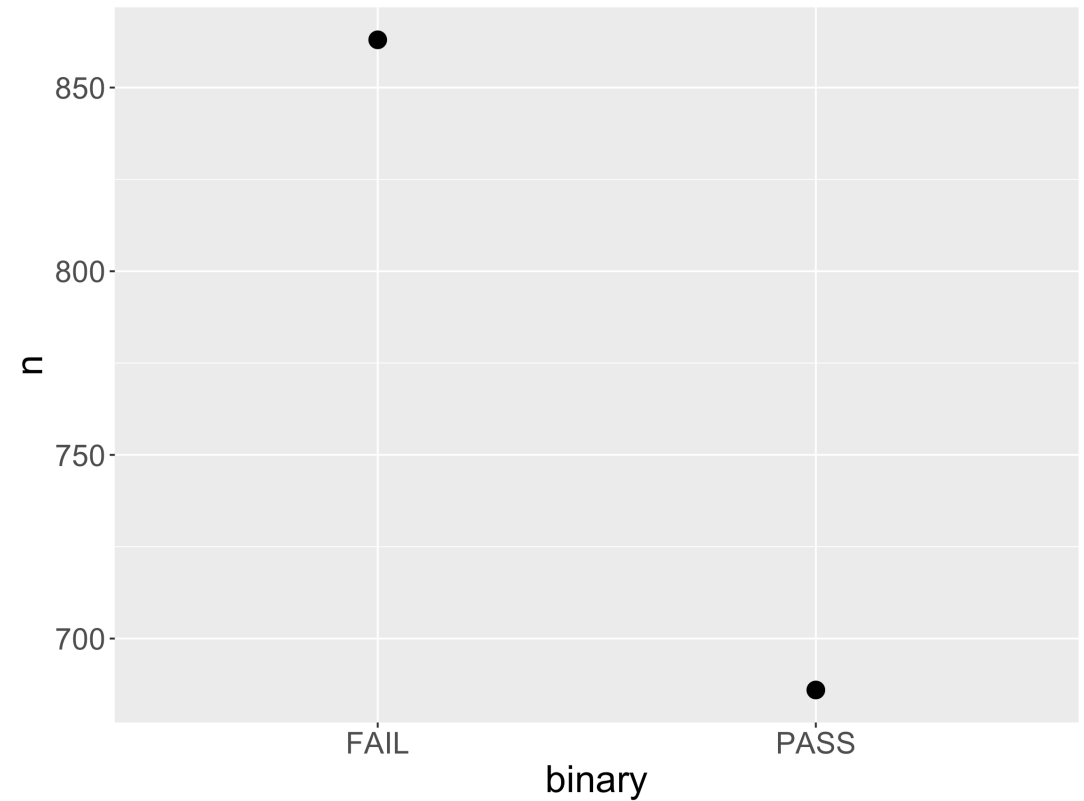
```
# A tibble: 2 × 2
  binary      n
  <chr>  <int>
1 FAIL    863
2 PASS    686
```

```
1 ggplot(data = movies_ag,
2       mapping = aes(x = binary, y = n)) +
3   geom_col()
```



geom_point again

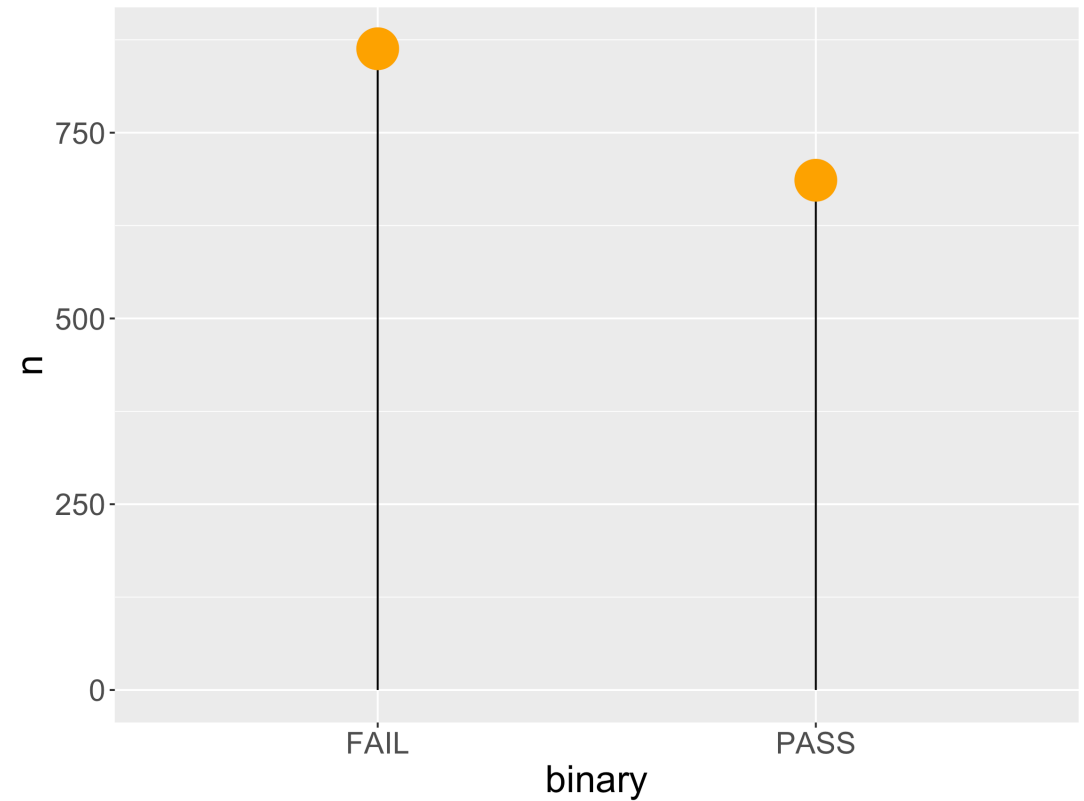
```
1 ggplot(data = movies_ag,  
2       mapping = aes(x = binary, y = n)) +  
3   geom_point(size = 4)
```



- If you are worried about the data-ink ratio...

geom_point + geom_segment

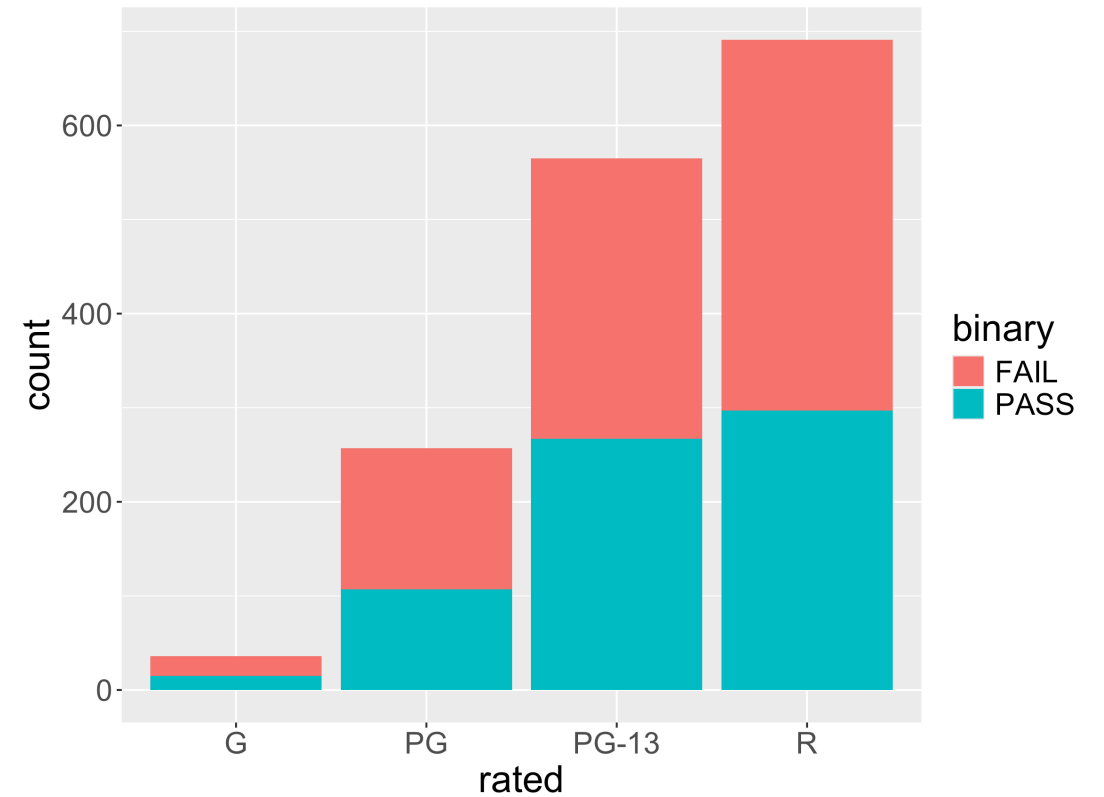
```
1 ggplot(data = movies_ag,  
2       mapping = aes(x = binary, y = n)) +  
3   geom_segment(mapping = aes(xend = binary),  
4               yend = 0) +  
5   geom_point(size = 10, color = "orange") +  
6   ylim(c(0, 875))
```



- Lollipop chart: compromise?

Two categorical variables: **geom_bar**

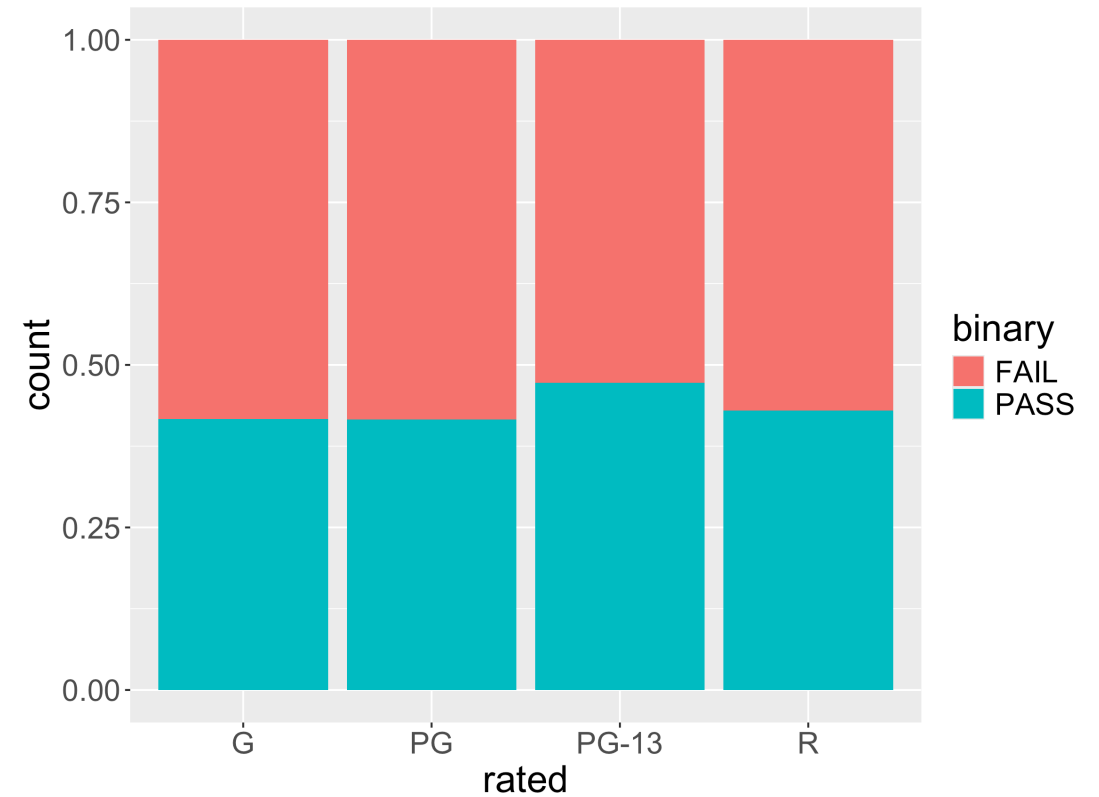
```
1 ggplot(data = movies,  
2       mapping = aes(x = rated,  
3                     fill = binary)) +  
4   geom_bar()
```



- Describe the mapping.

Two categorical variables: **geom_bar**

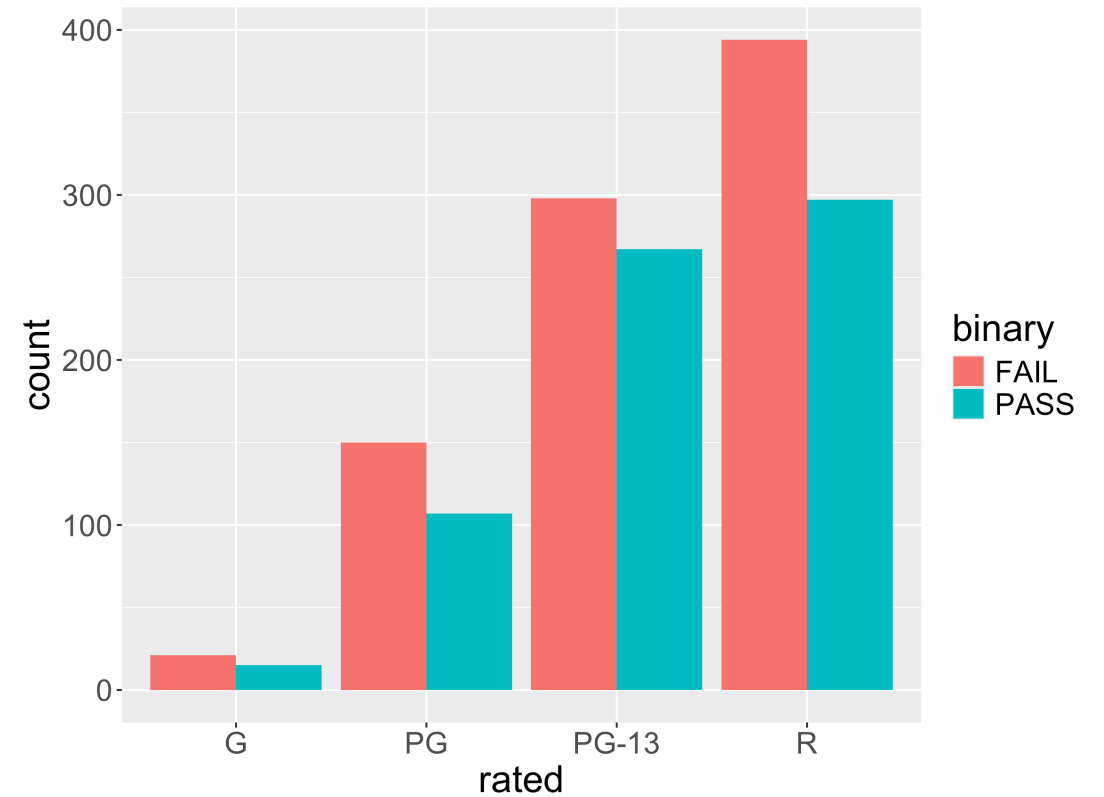
```
1 ggplot(data = movies,  
2       mapping = aes(x = rated,  
3                     fill = binary)) +  
4   geom_bar(position = "fill")
```



- Describe the mapping.

Two categorical variables: `geom_bar`

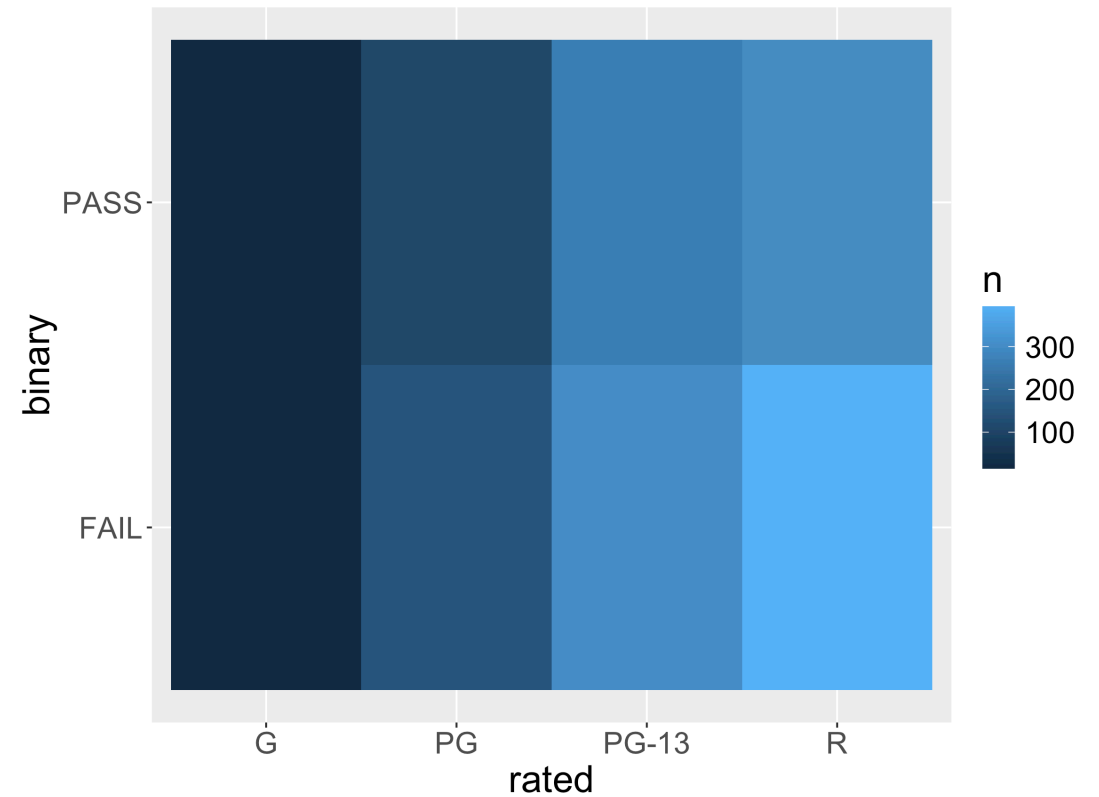
```
1 ggplot(data = movies,  
2       mapping = aes(x = rated,  
3                     fill = binary)) +  
4   geom_bar(position = "dodge")
```



- Describe the mapping.

Two categorical variables: `geom_tile`

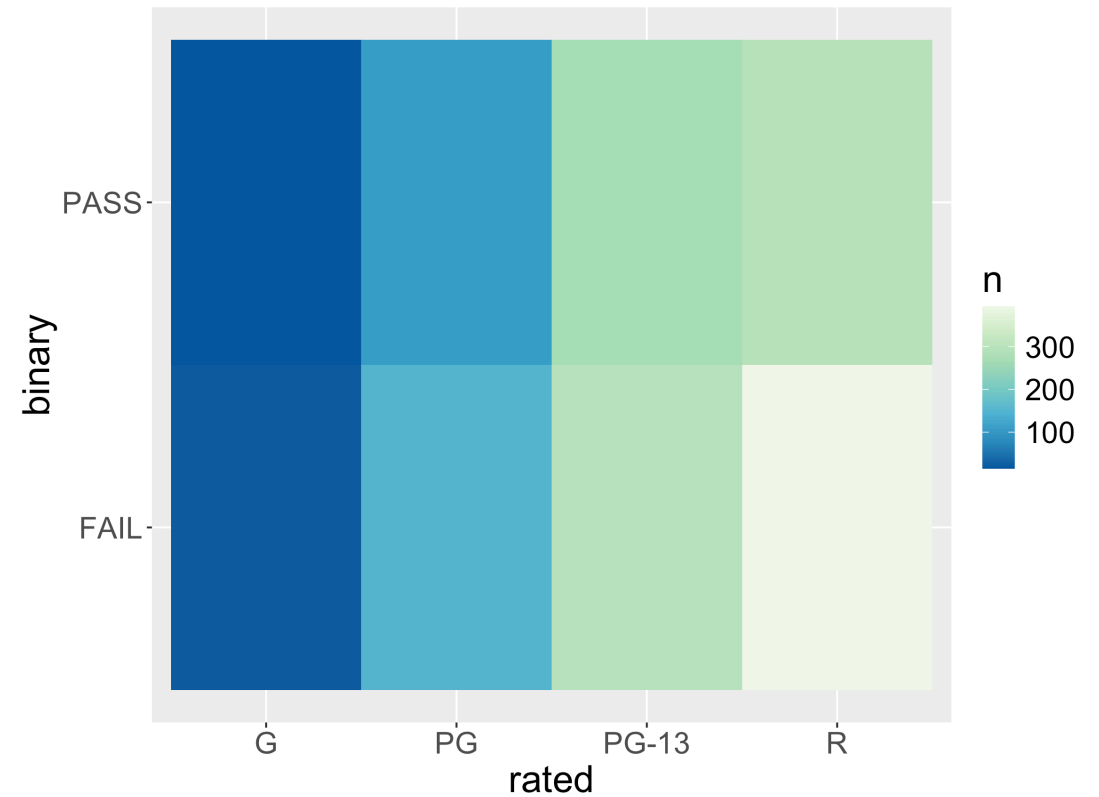
```
1 movies_ag <- count(movies, rated, binary)
2
3 ggplot(data = movies_ag,
4       mapping = aes(x = rated,
5                     y = binary,
6                     fill = n)) +
7   geom_tile()
```



- Describe the mapping.

Two categorical variables: `geom_tile`

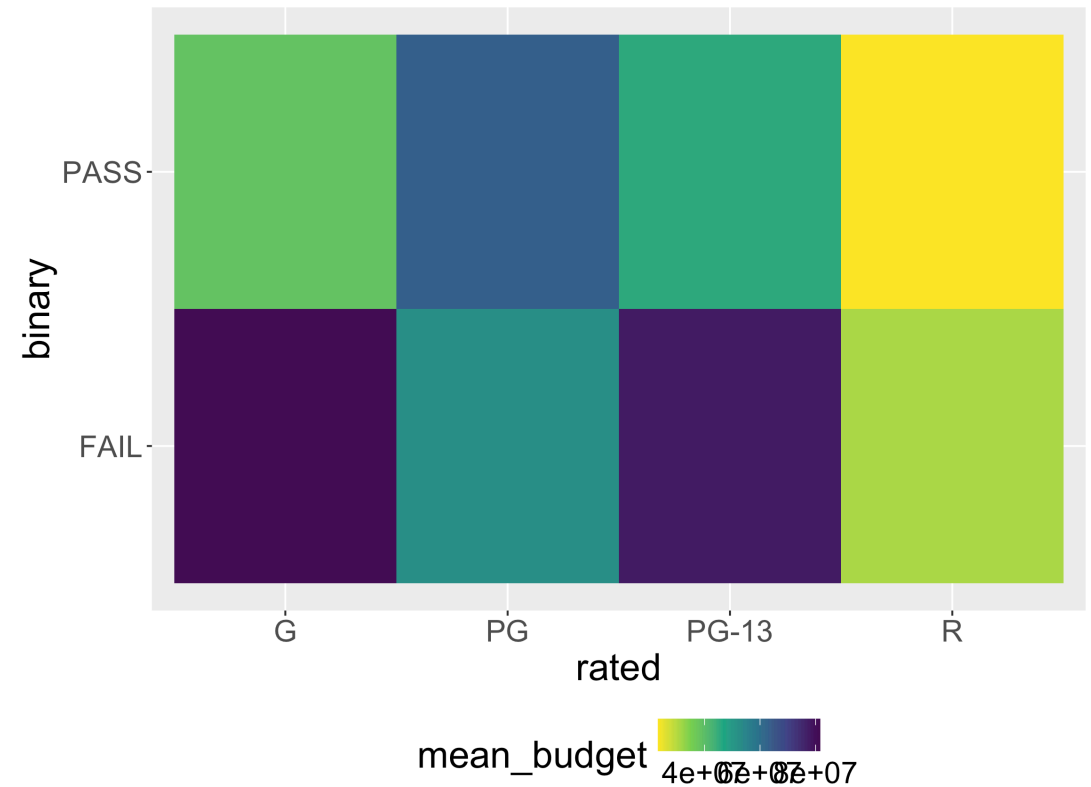
```
1 movies_ag <- count(movies, rated, binary)
2
3 ggplot(data = movies_ag,
4       mapping = aes(x = rated,
5                     y = binary,
6                     fill = n)) +
7   geom_tile() +
8   scale_fill_distiller(palette = 4)
```



- Change the `fill` scale!

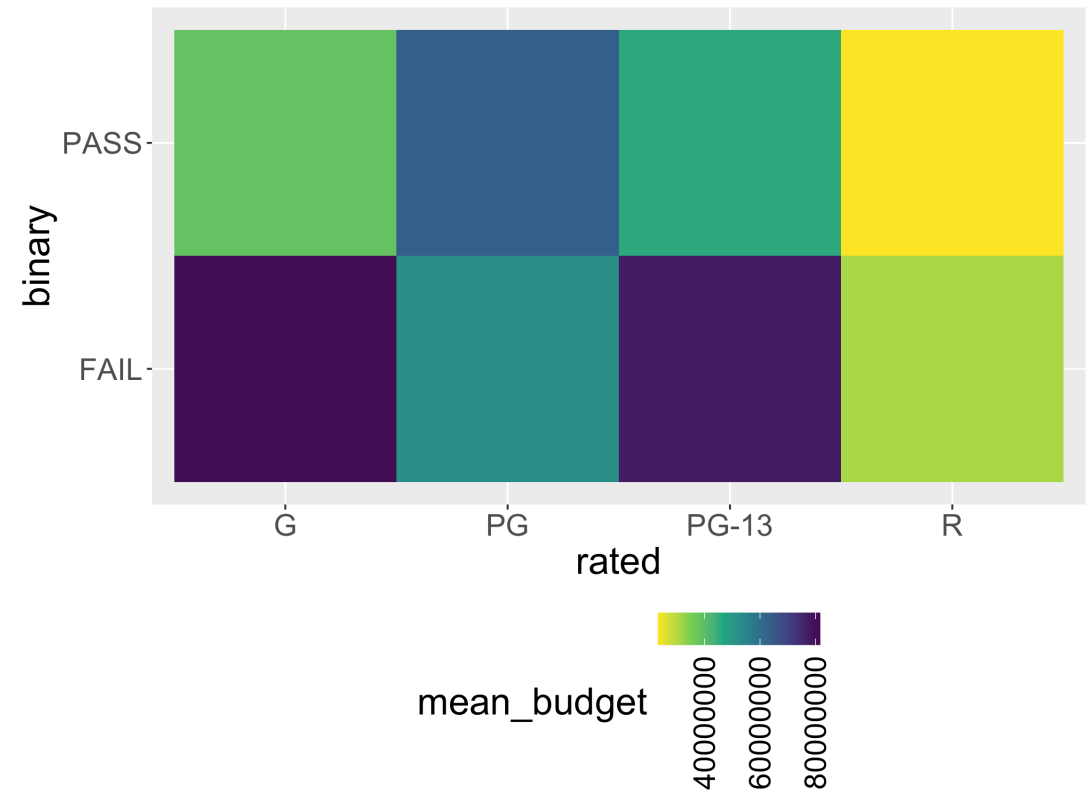
Can display more than frequencies!

```
1 movies_ag <- group_by(movies,  
2   rated, binary) %>%  
3   summarize(mean_budget = mean(budget))  
4  
5 ggplot(data = movies_ag,  
6   mapping = aes(x = rated,  
7   y = binary,  
8   fill = mean_budget)) +  
9   geom_tile() +  
10  scale_fill_viridis_c(direction = -1) +  
11  theme(legend.position = "bottom")
```



Can display more than frequencies!

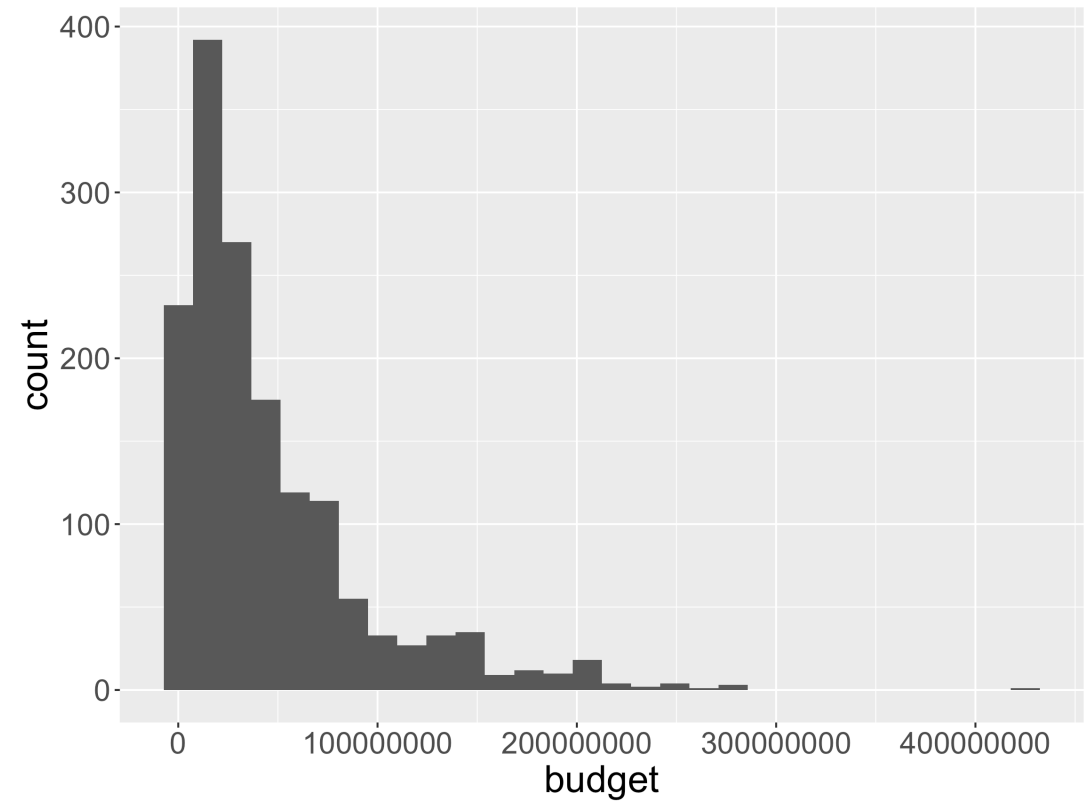
```
1 options(scipen = 999) # turn off scientific notation
2
3 movies_ag <- group_by(movies,
4                       rated, binary) %>%
5   summarize(mean_budget = mean(budget))
6
7 ggplot(data = movies_ag,
8       mapping = aes(x = rated,
9                     y = binary,
10                    fill = mean_budget)) +
11   geom_tile() +
12   scale_fill_viridis_c(direction = -1,
13                       guide = guide_colorbar(angle = 90))
14   theme(legend.position = "bottom")
```



What are useful **geoms** (graphs) for visualizing distributions?

Distributions: **geom_histogram**

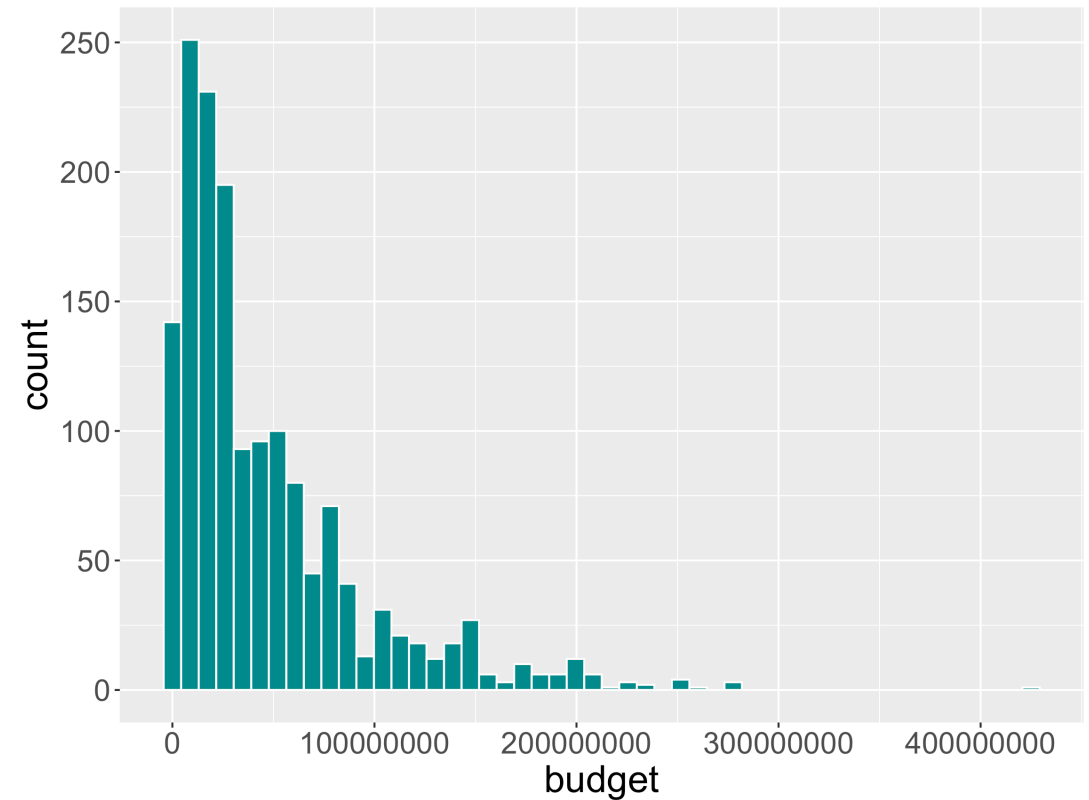
```
1 ggplot(movies, aes(x = budget)) +  
2   geom_histogram()
```



- Describe the mapping.

Distributions: `geom_histogram`

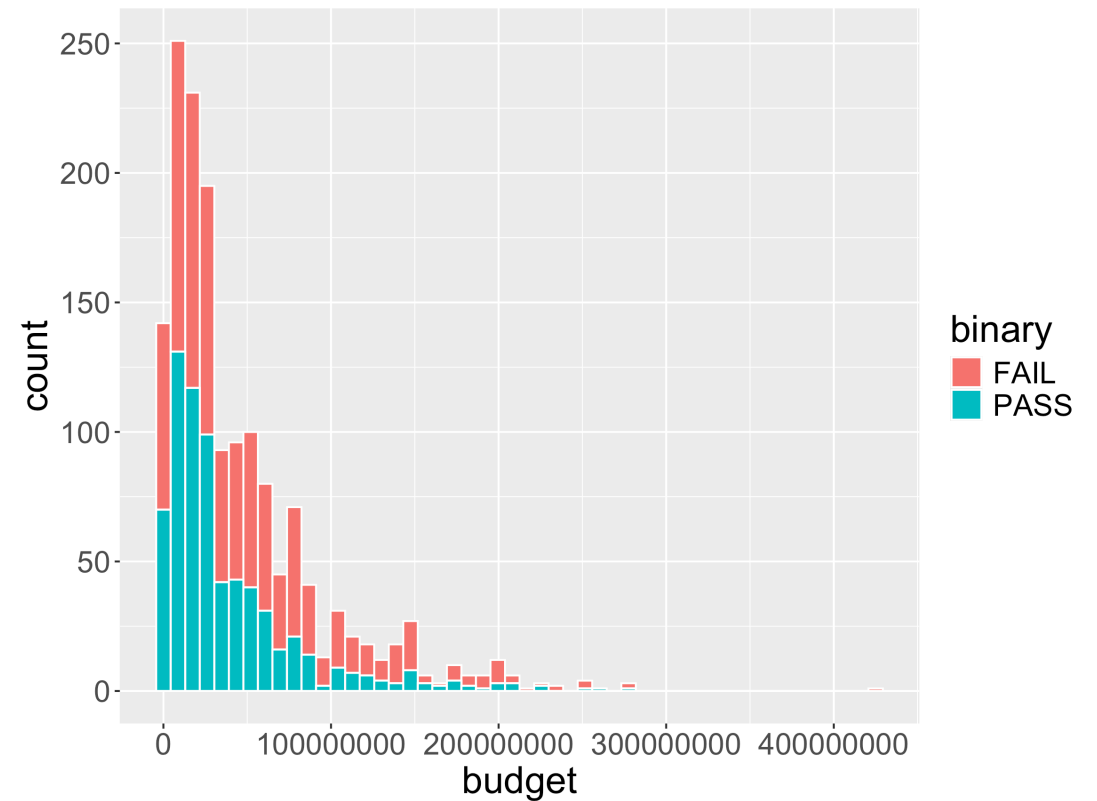
```
1 ggplot(movies, aes(x = budget)) +  
2   geom_histogram(bins = 50,  
3                 color = "white",  
4                 fill = "darkcyan")
```



- Can modify the mapping via the `binwidth` or `bins` arguments

Distributions: **geom_histogram**

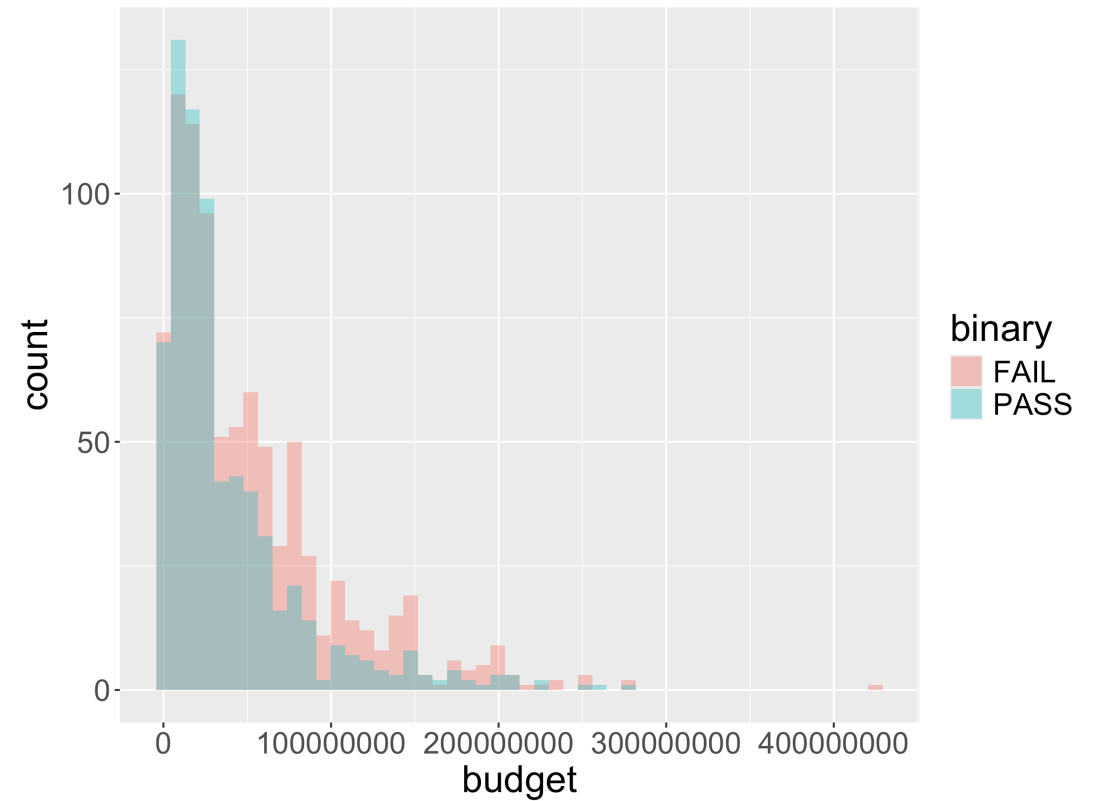
```
1 ggplot(movies, aes(x = budget,  
2                     fill = binary)) +  
3   geom_histogram(bins = 50,  
4                 color = "white")
```



- What is problematic about this graph?

Distributions: **geom_histogram**

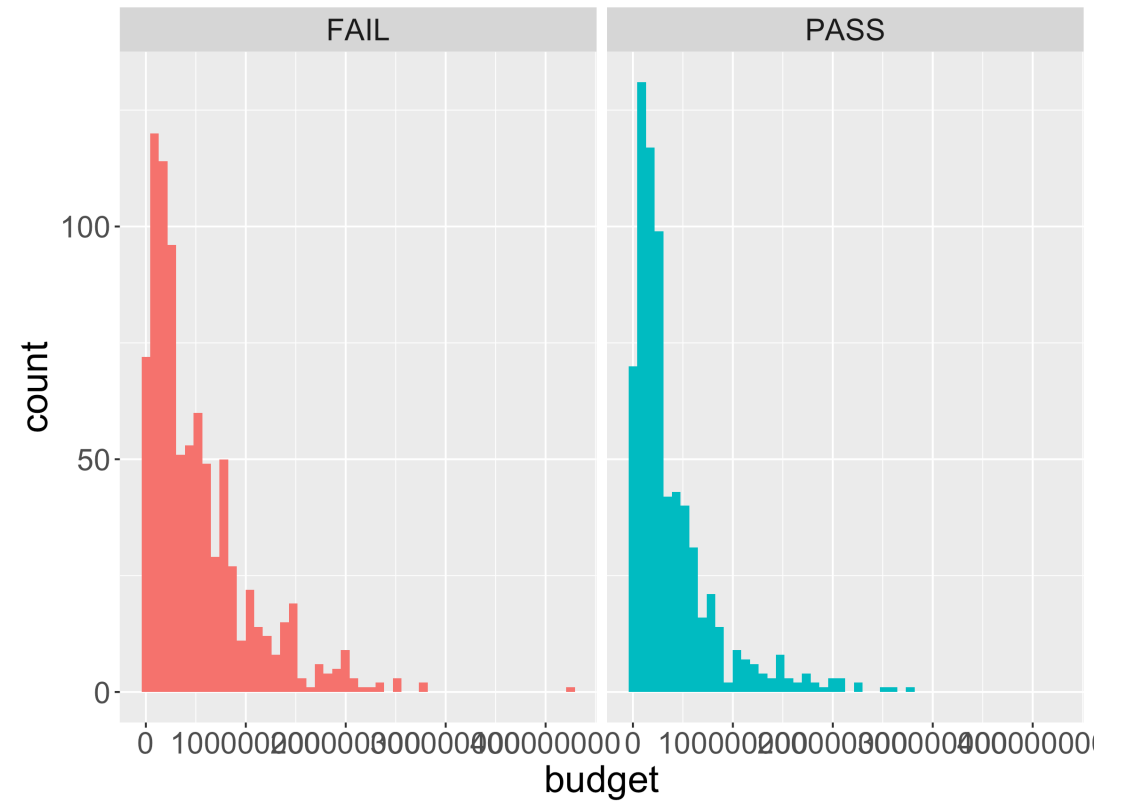
```
1 ggplot(movies, aes(x = budget,  
2                   fill = binary)) +  
3   geom_histogram(bins = 50,  
4                 alpha = 0.4,  
5                 position = "identity")
```



- Still problematic.

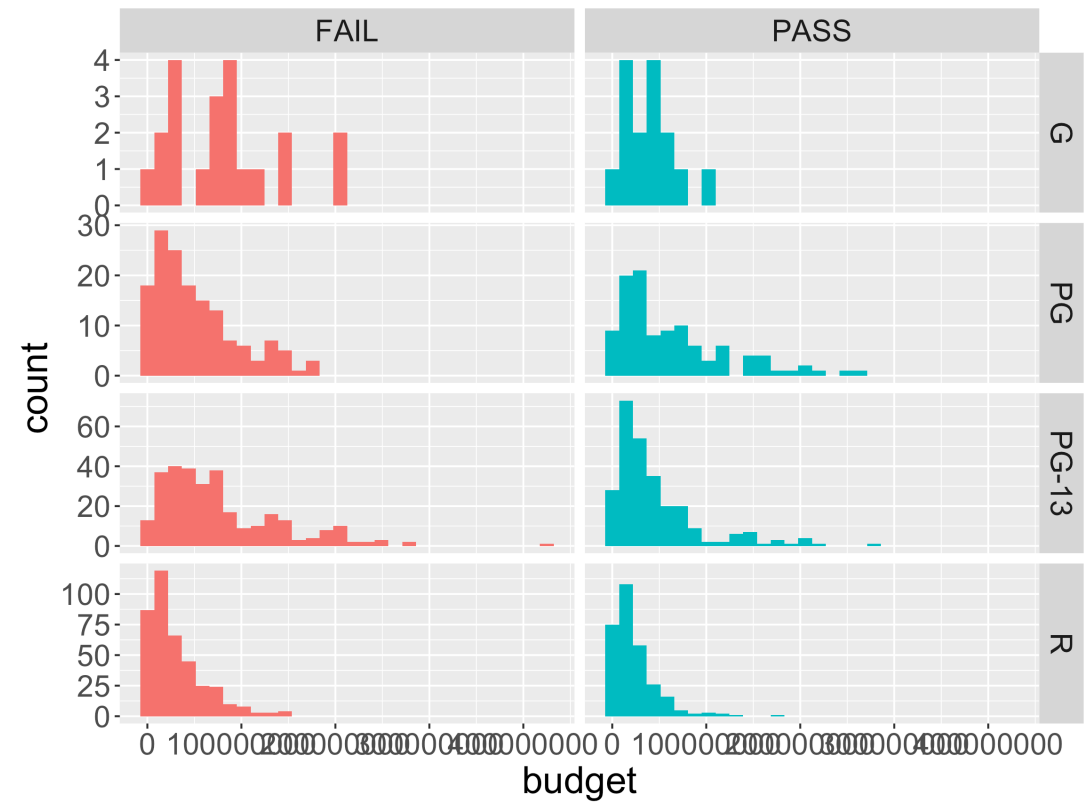
One option: Faceting

```
1 ggplot(movies, aes(x = budget,  
2                   fill = binary)) +  
3   geom_histogram(bins = 50) +  
4   facet_wrap(~binary) +  
5   guides(fill = "none")
```



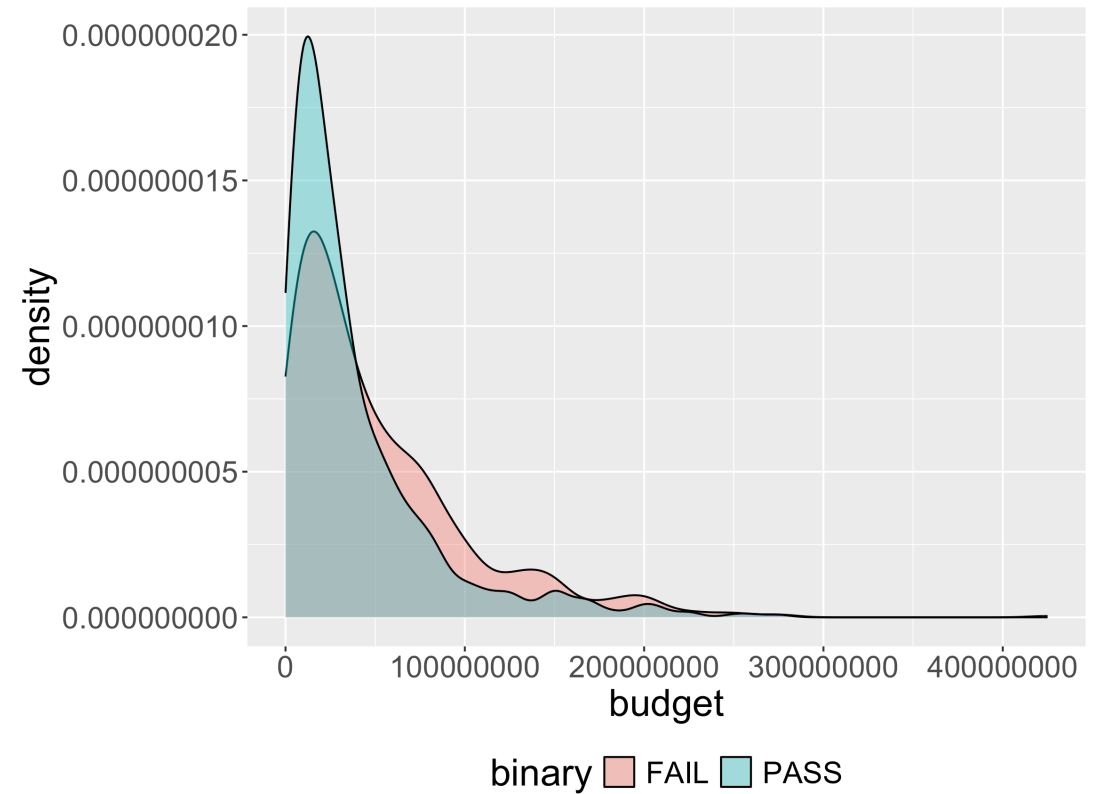
One option: Faceting

```
1 ggplot(movies, aes(x = budget,  
2                   fill = binary)) +  
3   geom_histogram() +  
4   facet_grid(rated ~ binary, scales = "free_y") +  
5   guides(fill = "none")
```



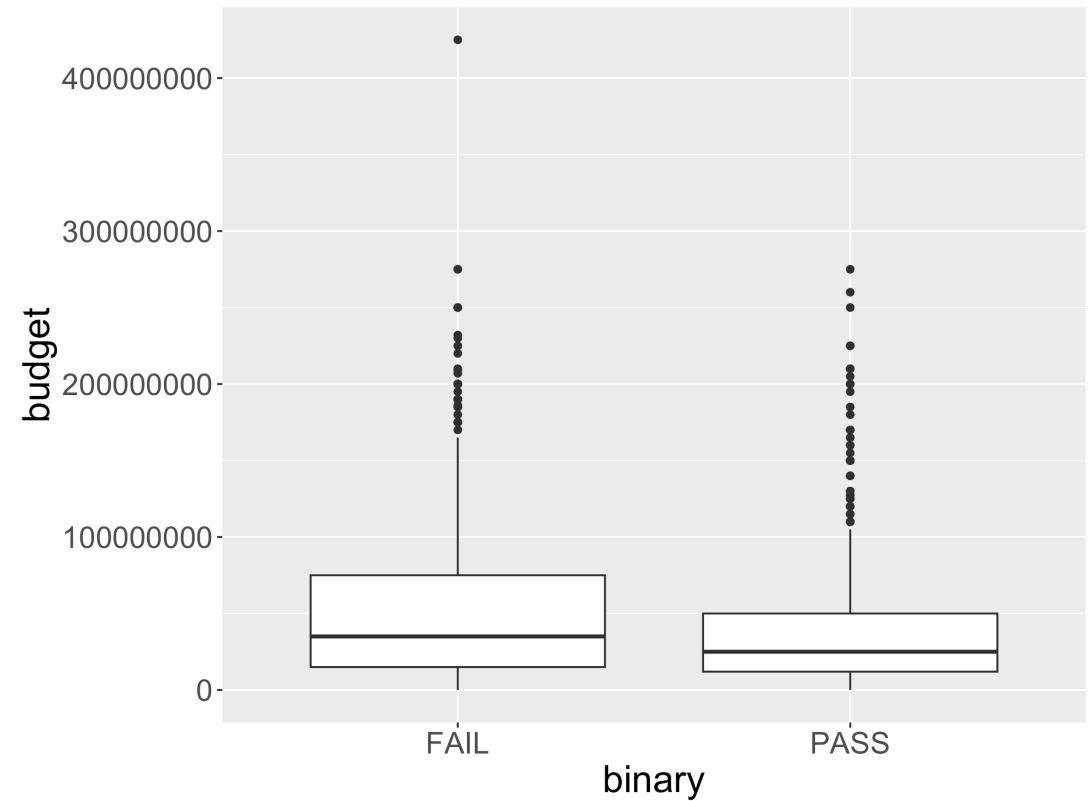
Another option: **geom_density**

```
1 ggplot(movies, aes(x = budget,  
2                   fill = binary)) +  
3   geom_density(alpha = 0.4) +  
4   theme(legend.position = "bottom")
```



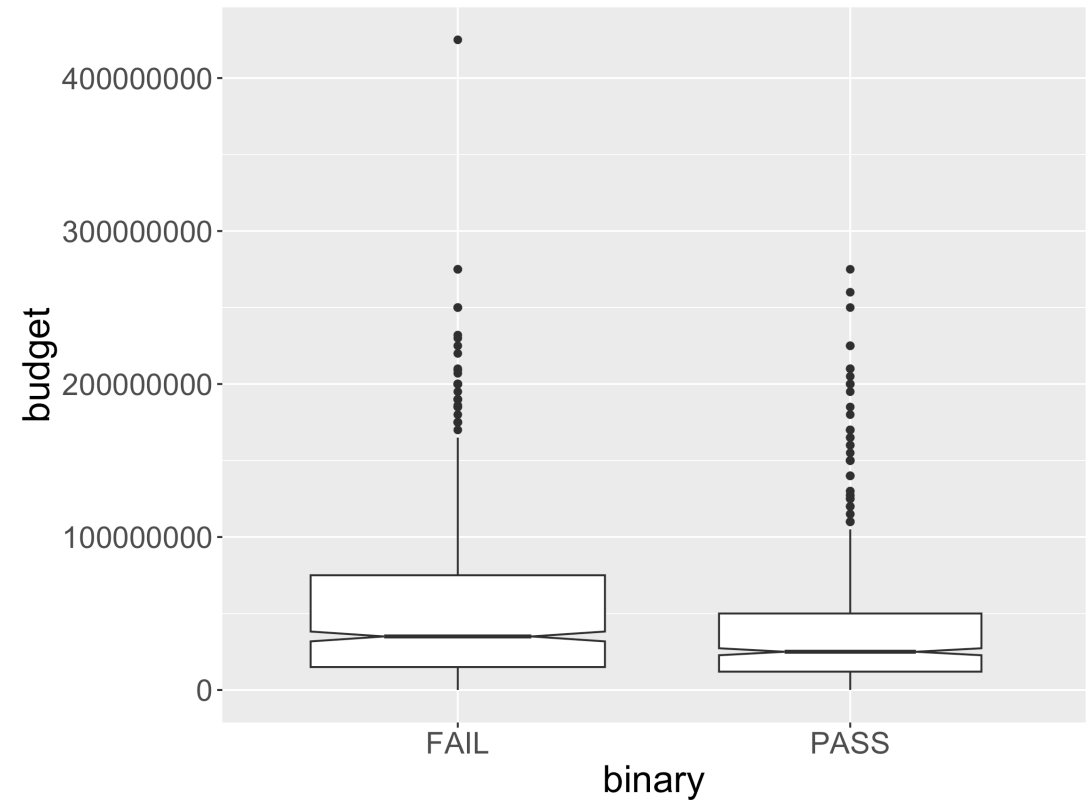
Distributions: **geom_boxplot**

```
1 ggplot(movies, aes(x = binary,  
2                   y = budget)) +  
3   geom_boxplot()
```



Distributions: `geom_boxplot`

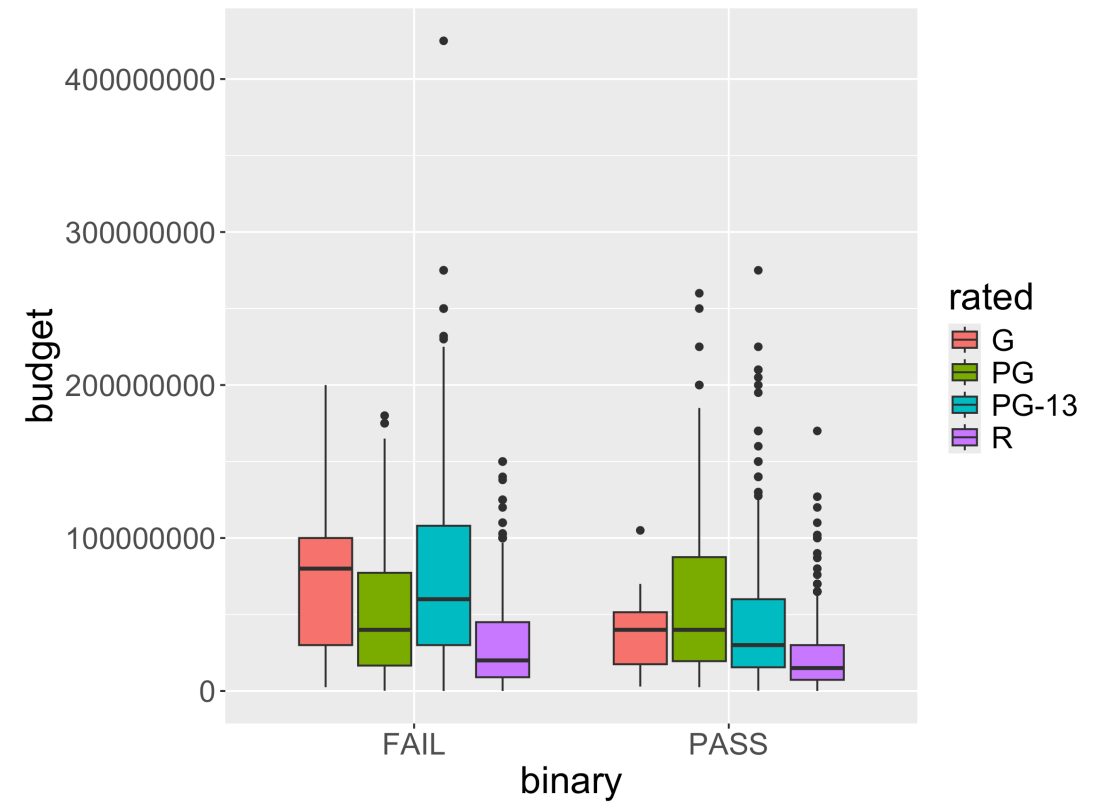
```
1 ggplot(movies, aes(x = binary,  
2                   y = budget)) +  
3   geom_boxplot(varwidth = TRUE,  
4               notch = TRUE)
```



- What does `varwidth` do?
- Why might we add `notch = TRUE`?

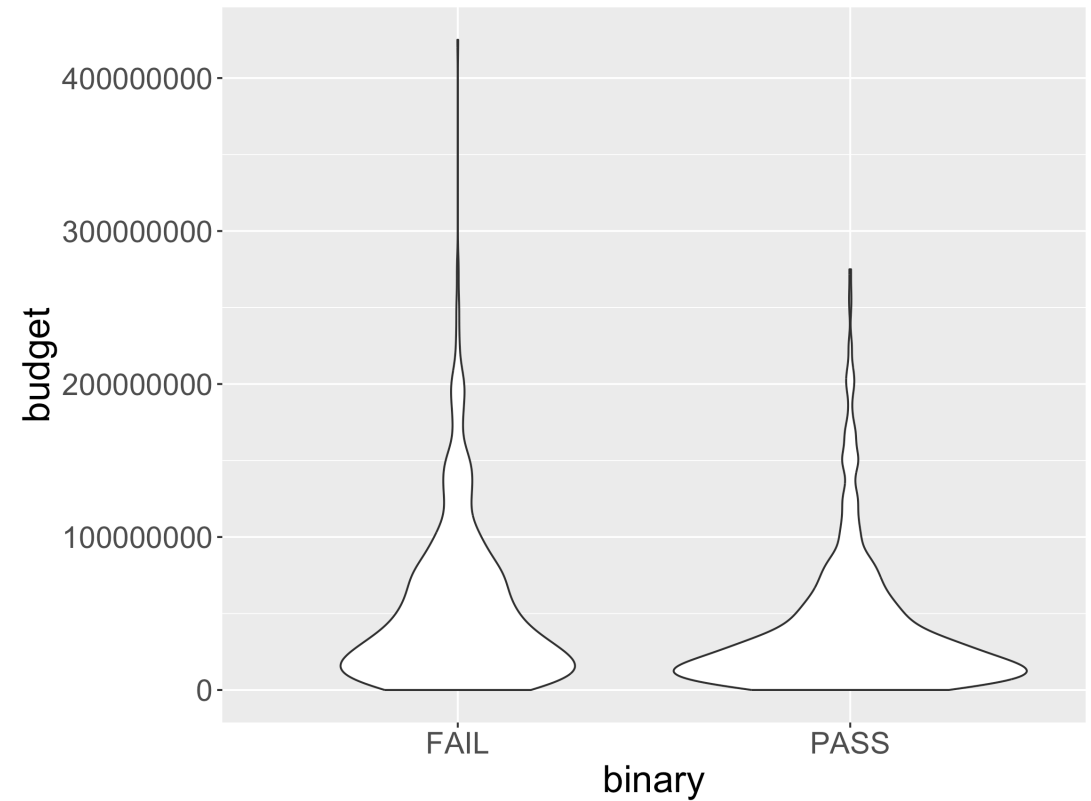
Distributions: `geom_boxplot`

```
1 ggplot(movies, aes(x = binary,  
2                   y = budget,  
3                   fill = rated)) +  
4   geom_boxplot()
```



Distributions: **geom_violin**

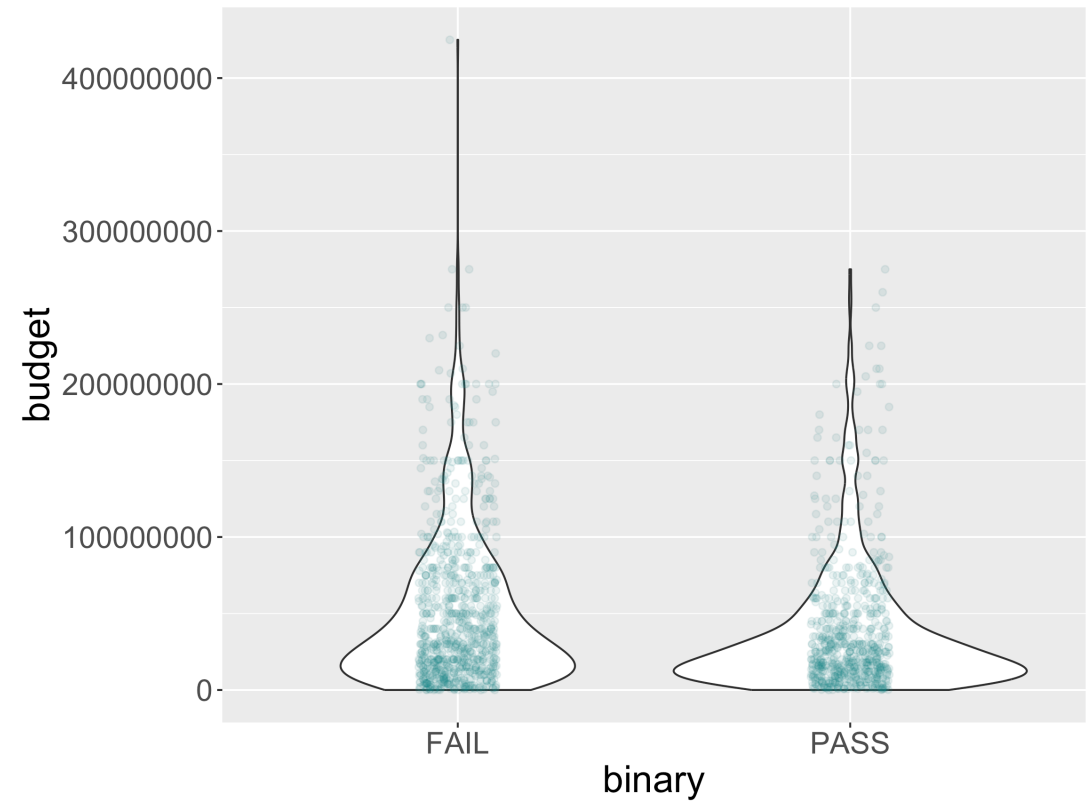
```
1 ggplot(movies, aes(x = binary,  
2                   y = budget)) +  
3   geom_violin()
```



- Utility of the violin over the box?

Distributions: **geom_violin**

```
1 ggplot(movies, aes(x = binary,  
2                   y = budget)) +  
3   geom_violin() +  
4   geom_jitter(alpha = .1,  
5               width = .1,  
6               color = "darkcyan")
```



Reminders

- Office Hours Schedule
- P-Set 1 released at 9am on Thursday.
 - Will discuss how to access the p-sets through GitHub on Wednesday.